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Okazaki et al.

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(54) **SHEET CONVEYANCE APPARATUS, SHEET SUPPORT APPARATUS, AND IMAGE FORMING APPARATUS**

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2402/45 (2013.01); **B65H 2404/67** (2013.01);
B65H 2801/06 (2013.01)

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2405/324; **B65H 2402/45**; **B65H**
2402/412

See application file for complete search history.

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(57) **ABSTRACT**

A sheet conveyance apparatus includes an apparatus body including a conveyance assembly to convey a sheet, and a door pivotally supported around a pivot axis disposed in a lower portion of the apparatus body such that the door is opened and closed with respect to the apparatus body. The apparatus body includes a handle assembly including a first holder used to lift the apparatus body and is disposed inside a width of the door in an axis direction of the pivot axis. The handle assembly includes a first engaging section and a second engaging section disposed on the pivot axis, wherein the door includes a first engaged section to engage with the first engaging section and a second engaged section to engage with the second engaging section. The door is pivotably supported around the first engaging section and the second engaging section with respect to the handle assembly.

13 Claims, 17 Drawing Sheets

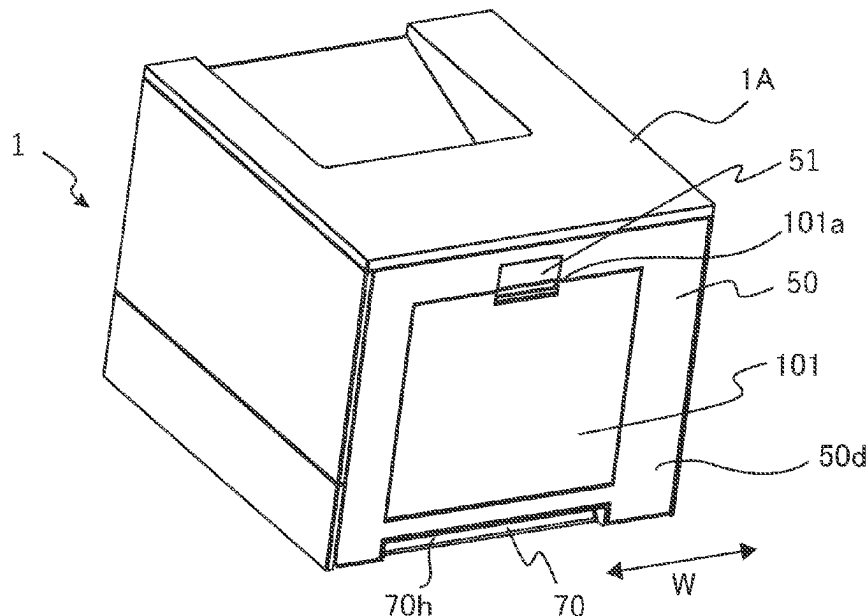


FIG.1

$\frac{1}{5}$

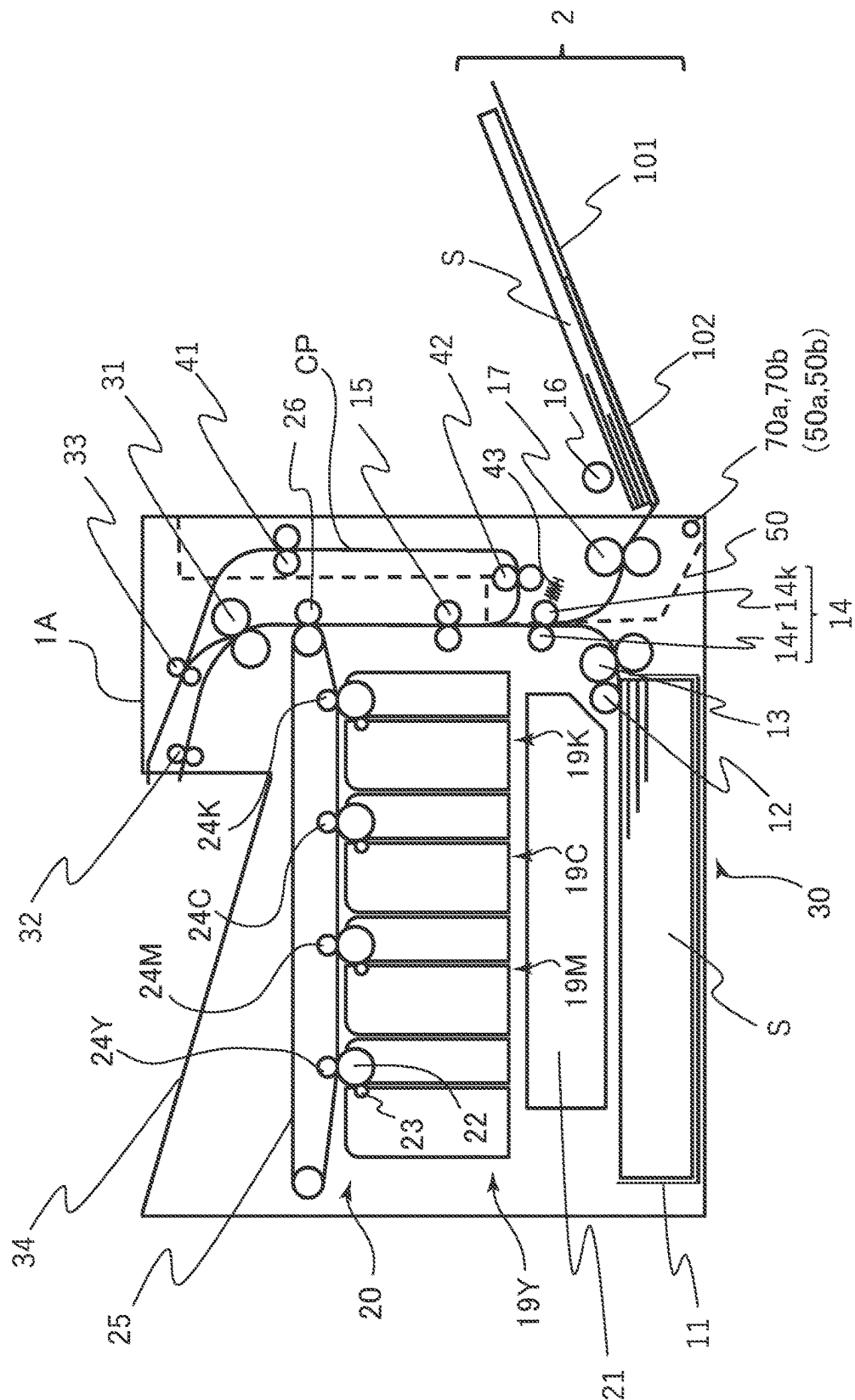


FIG.2A

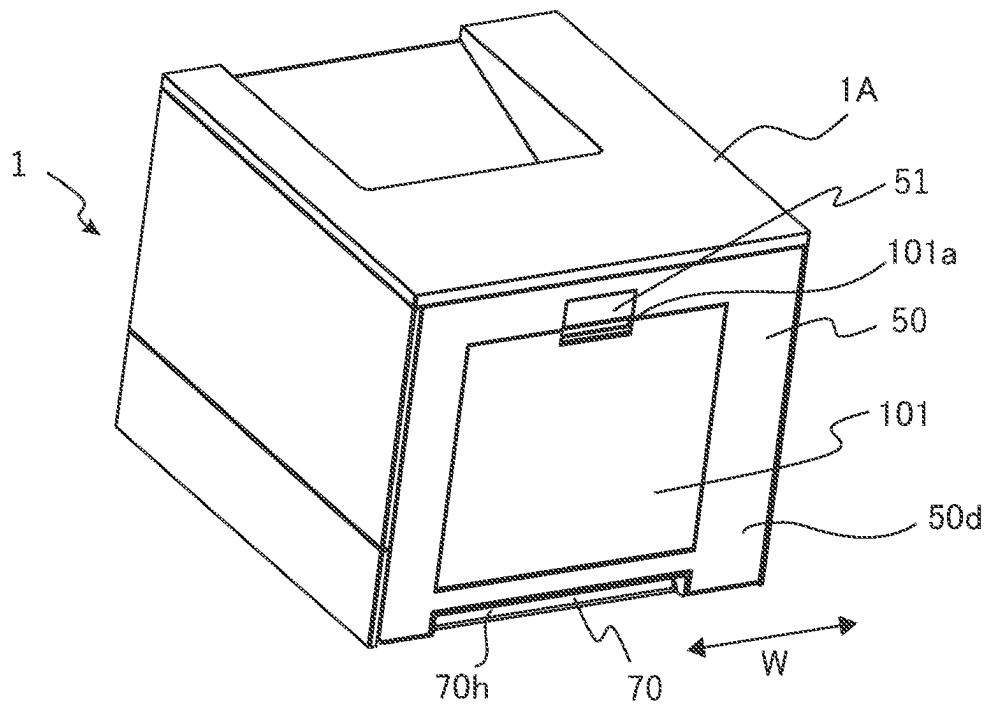


FIG.2B

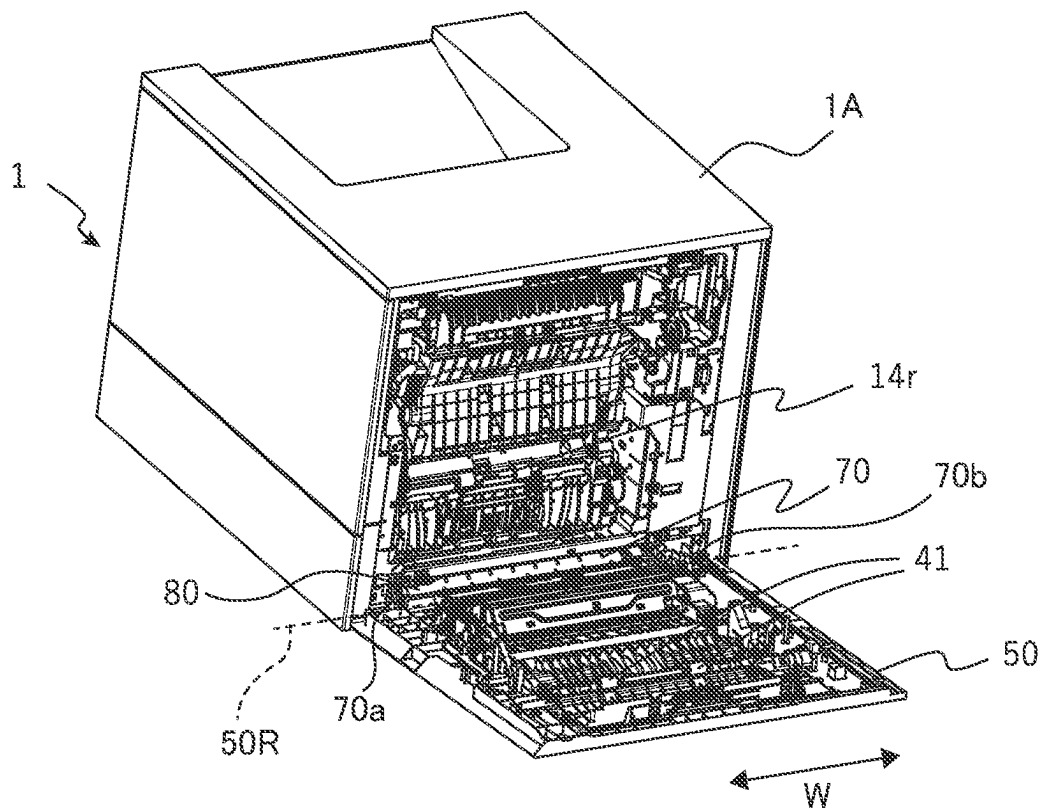


FIG.3

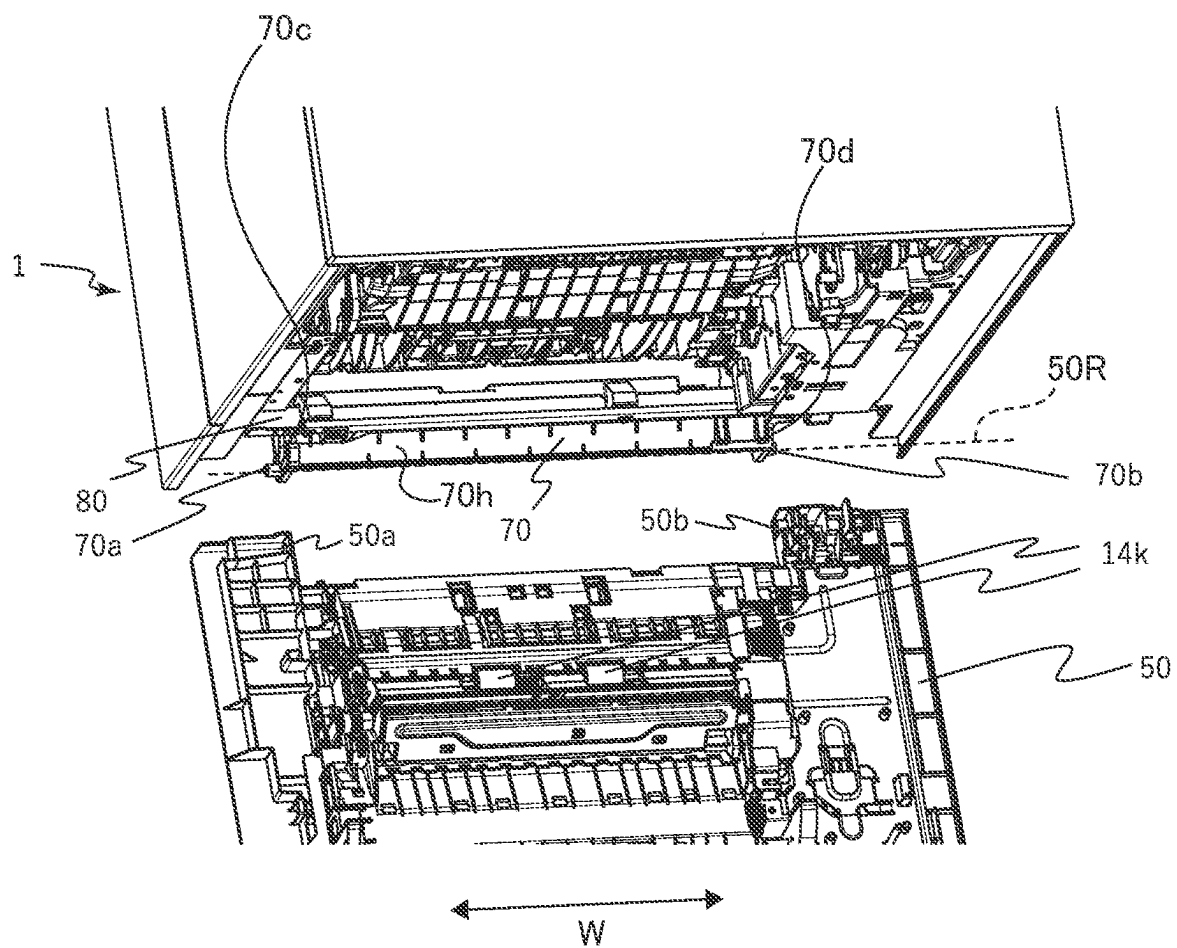


FIG.4A

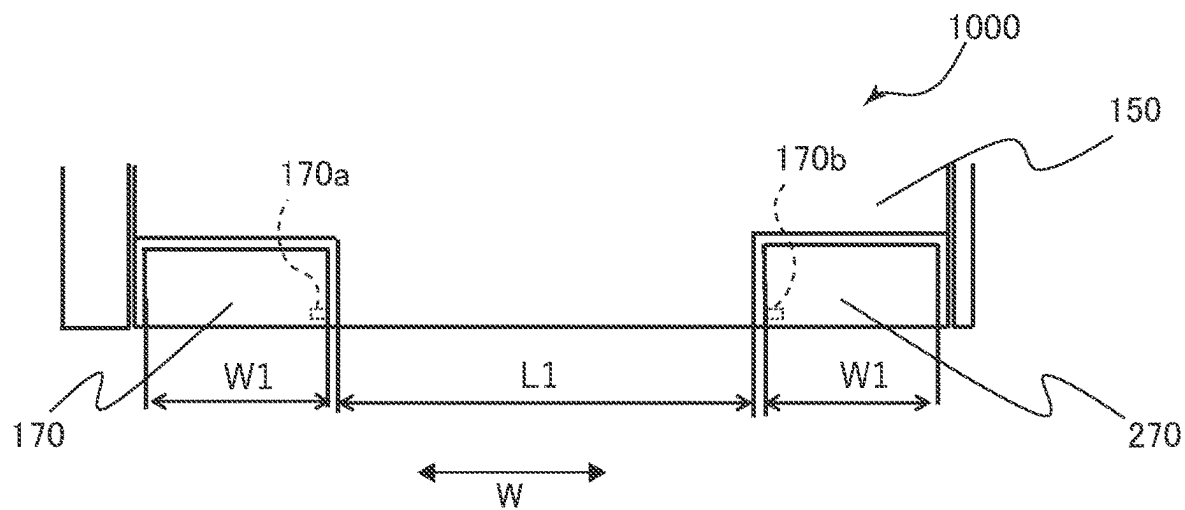


FIG.4B

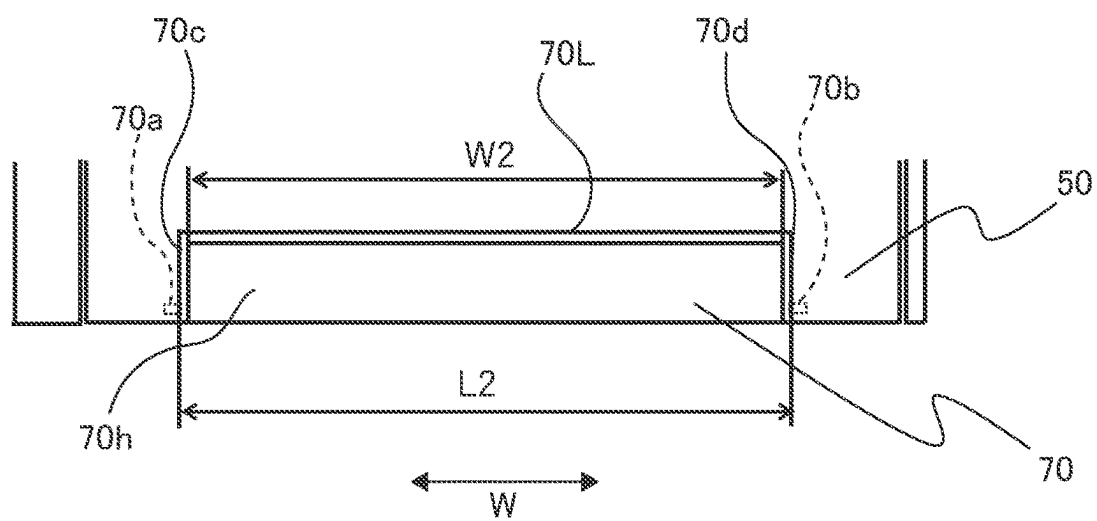


FIG.5

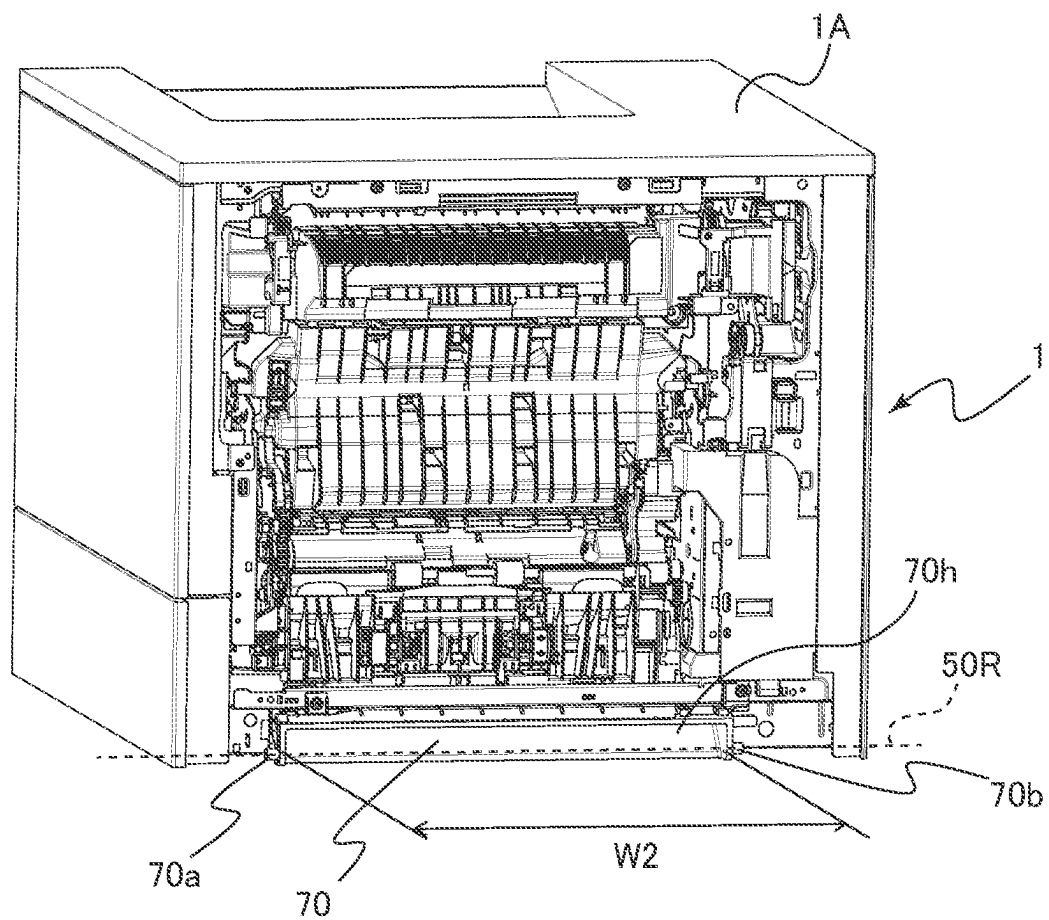


FIG. 6

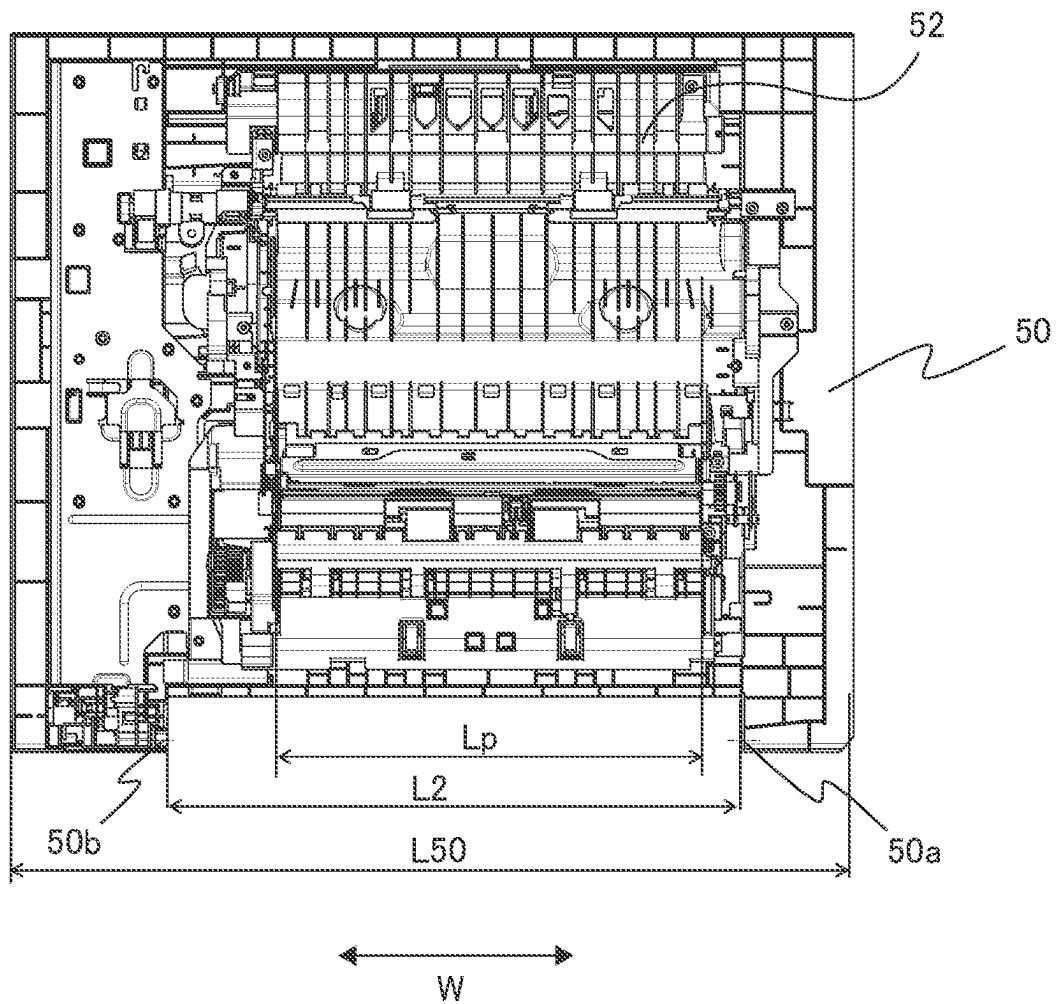


FIG. 7

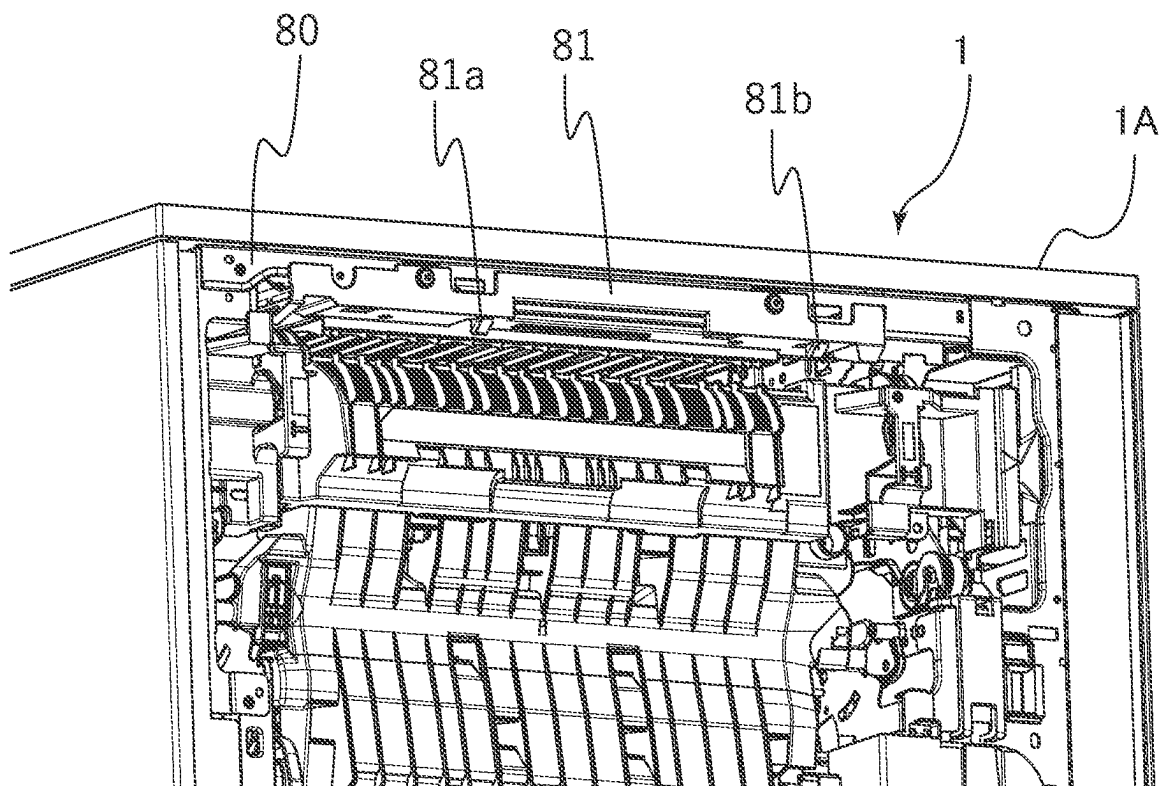


FIG.8

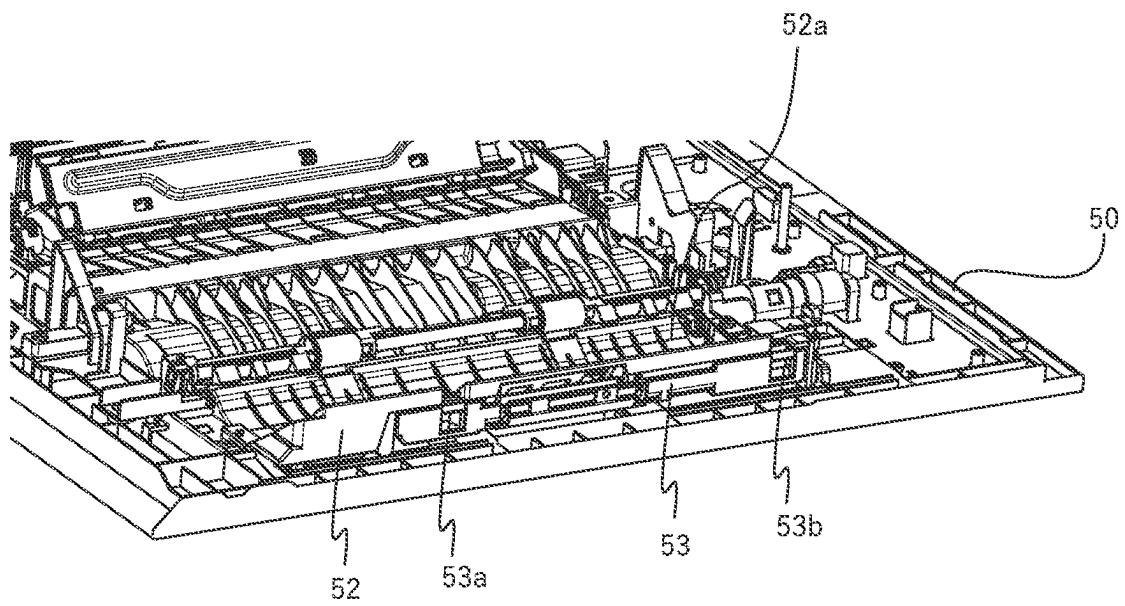


FIG. 9A

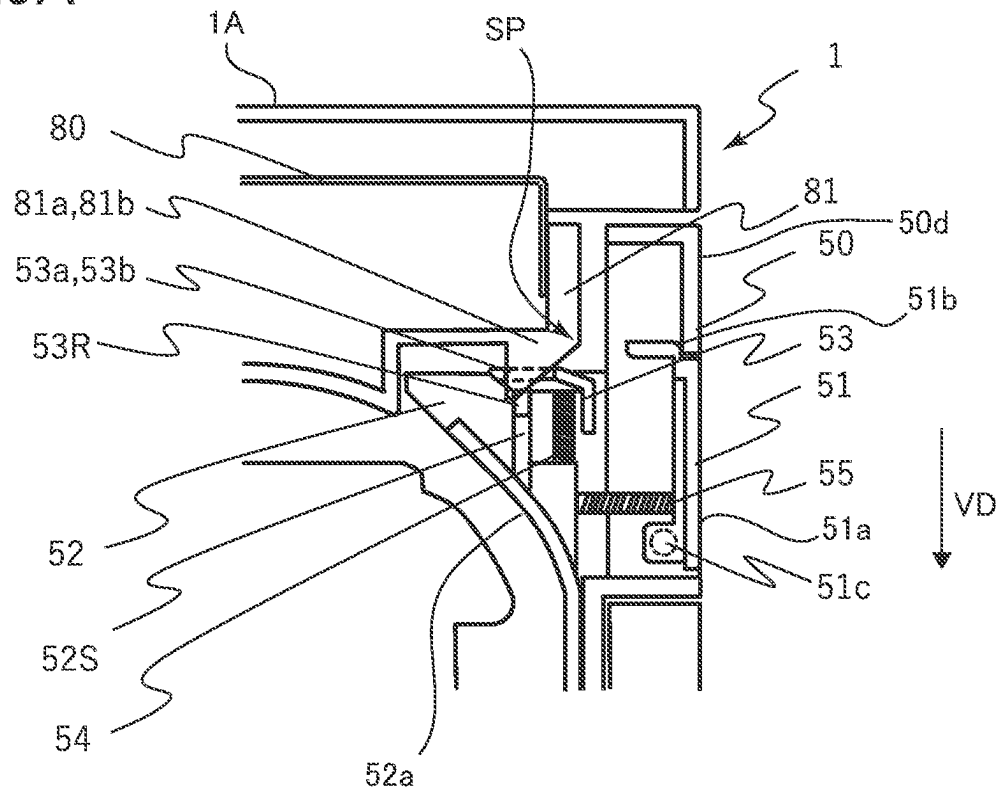


FIG. 9B

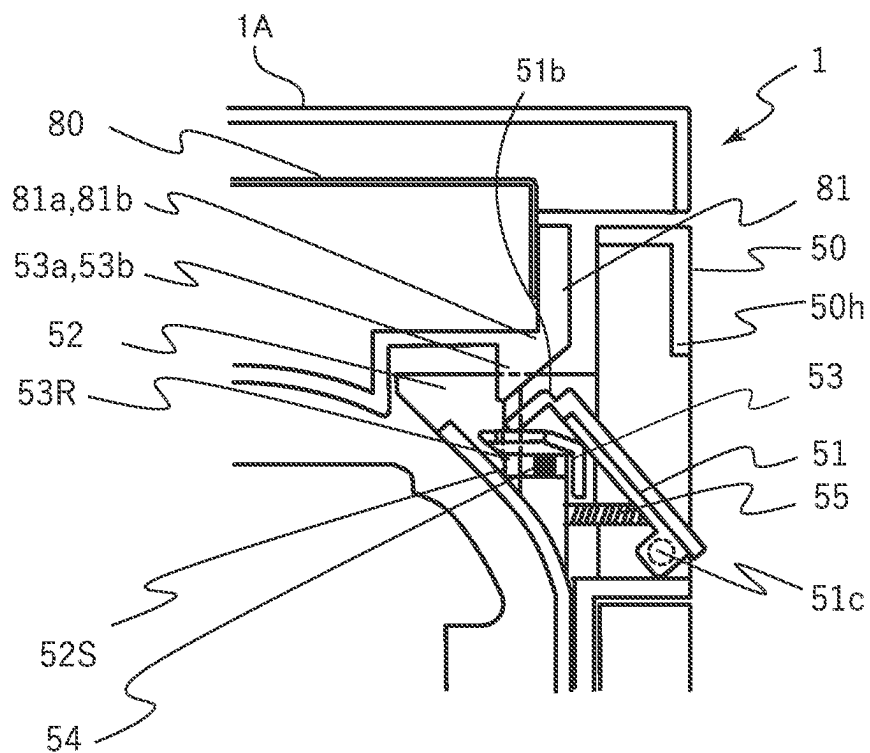


FIG.10A

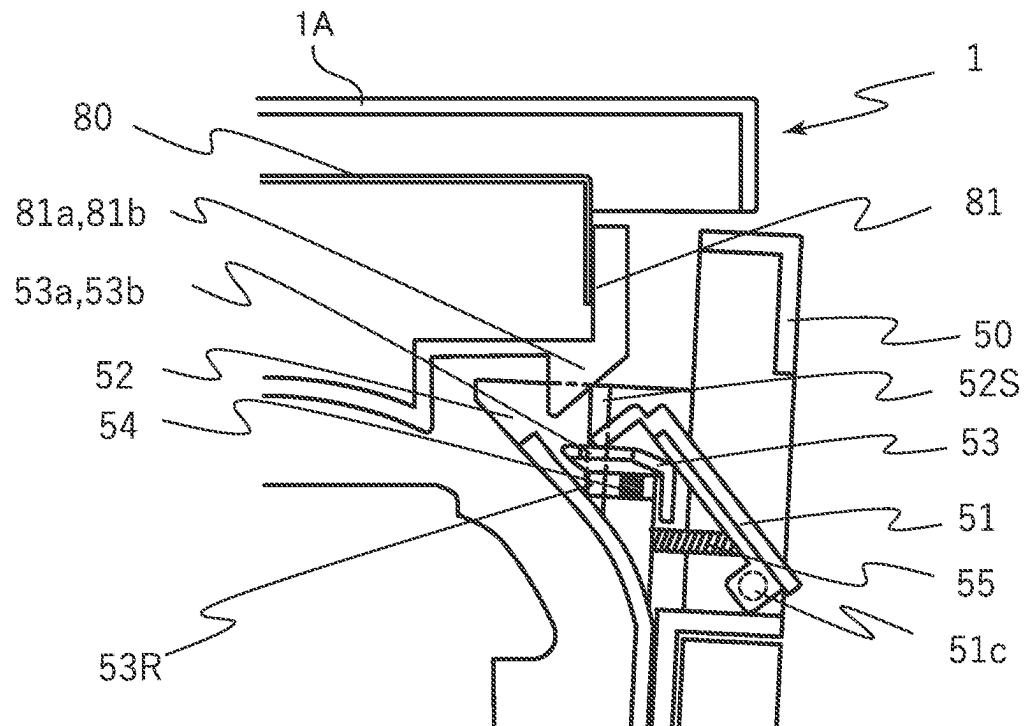


FIG.10B

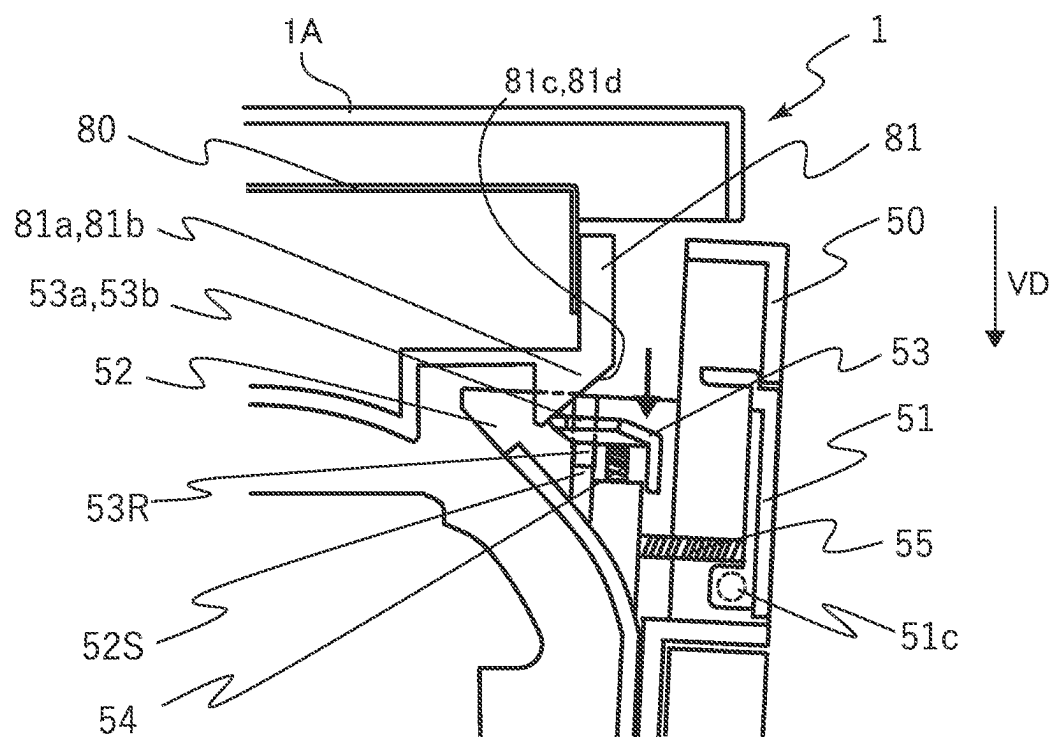


FIG. 11

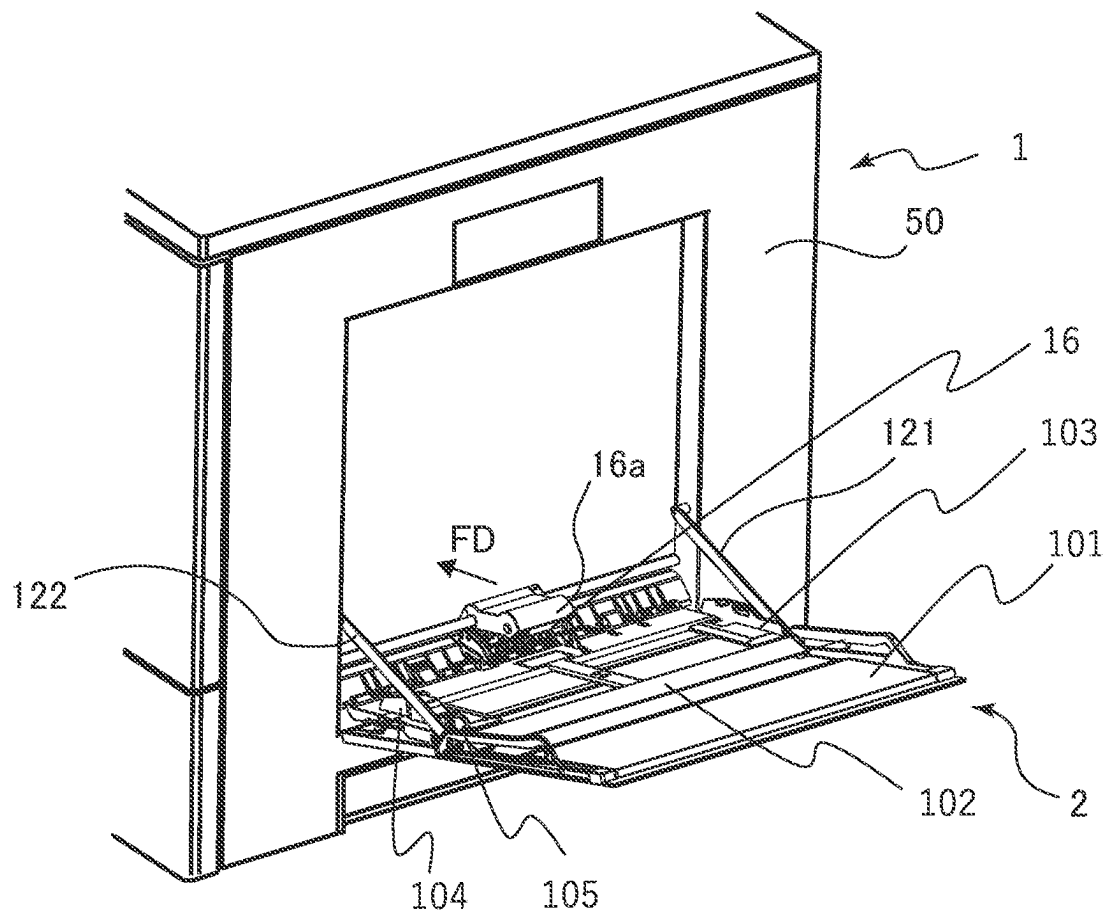


FIG.12

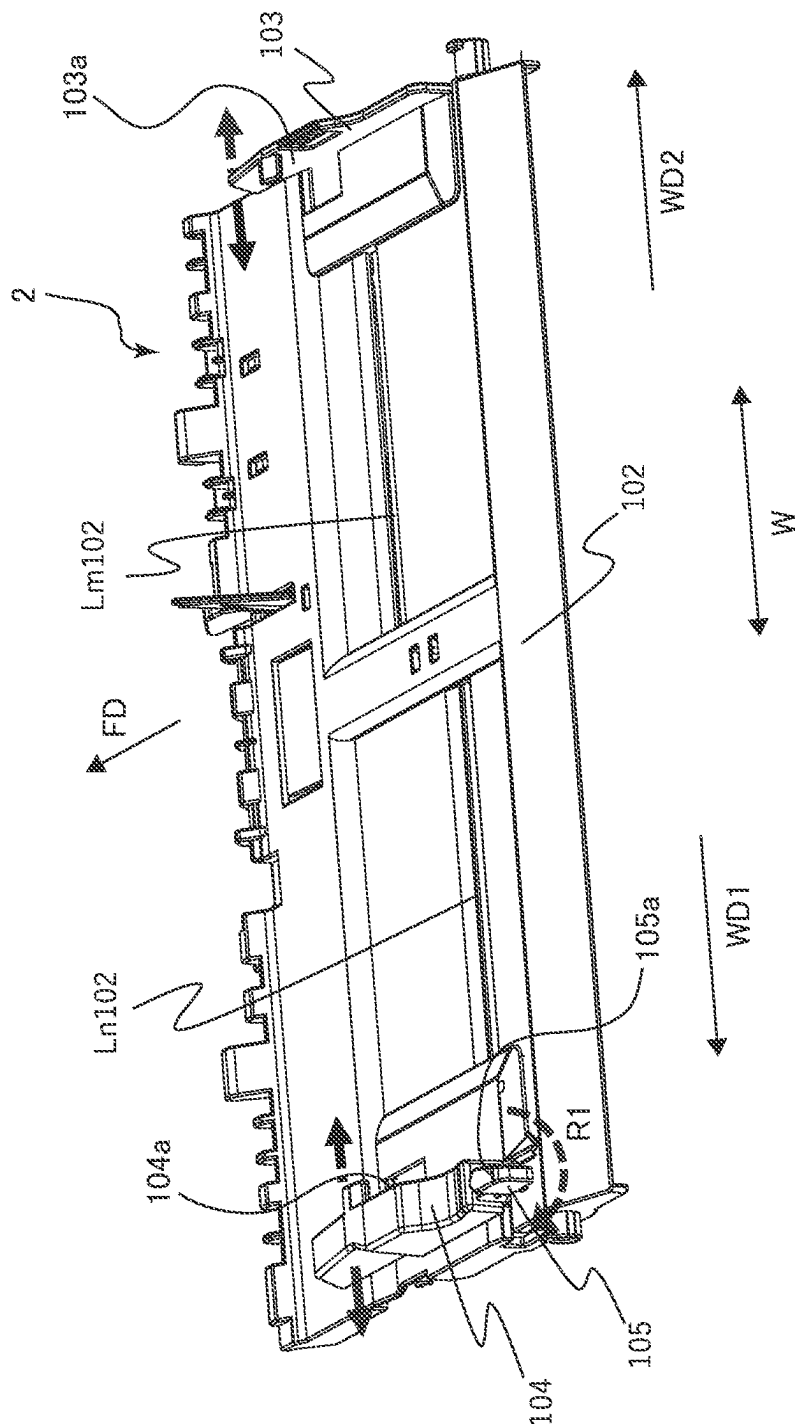


FIG. 13

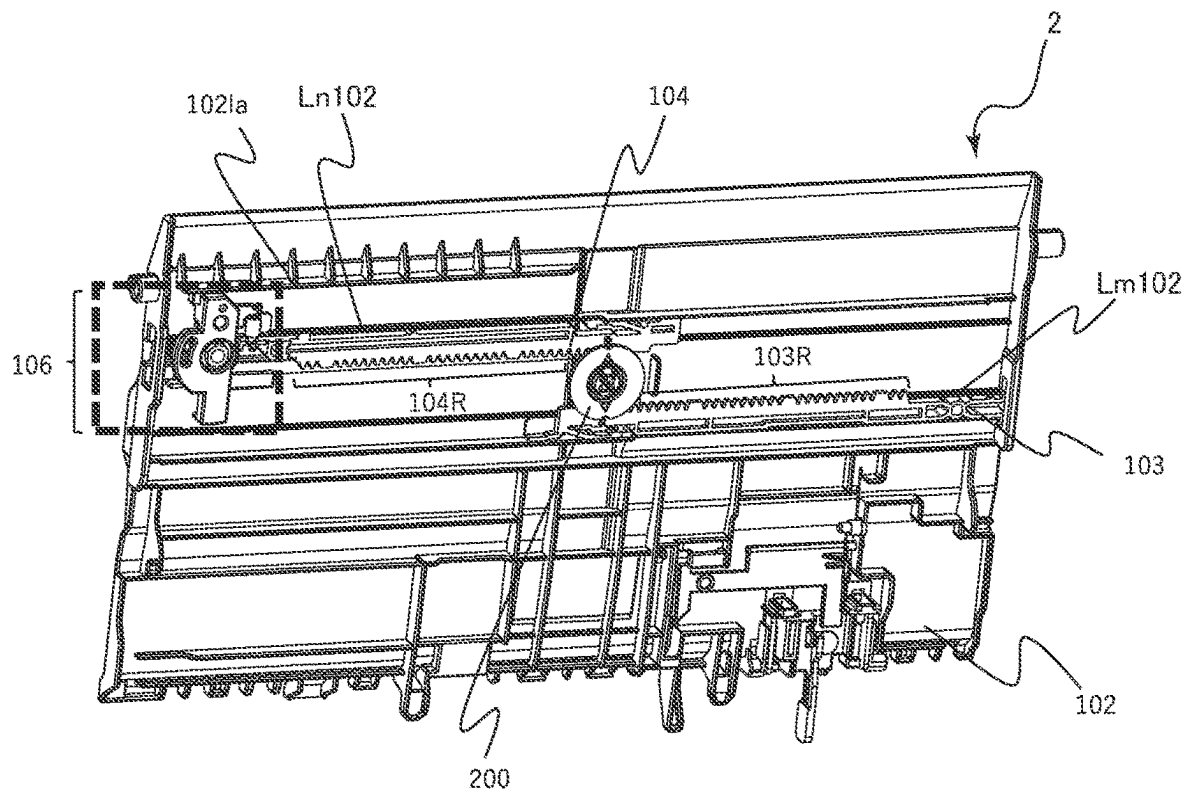


FIG.14

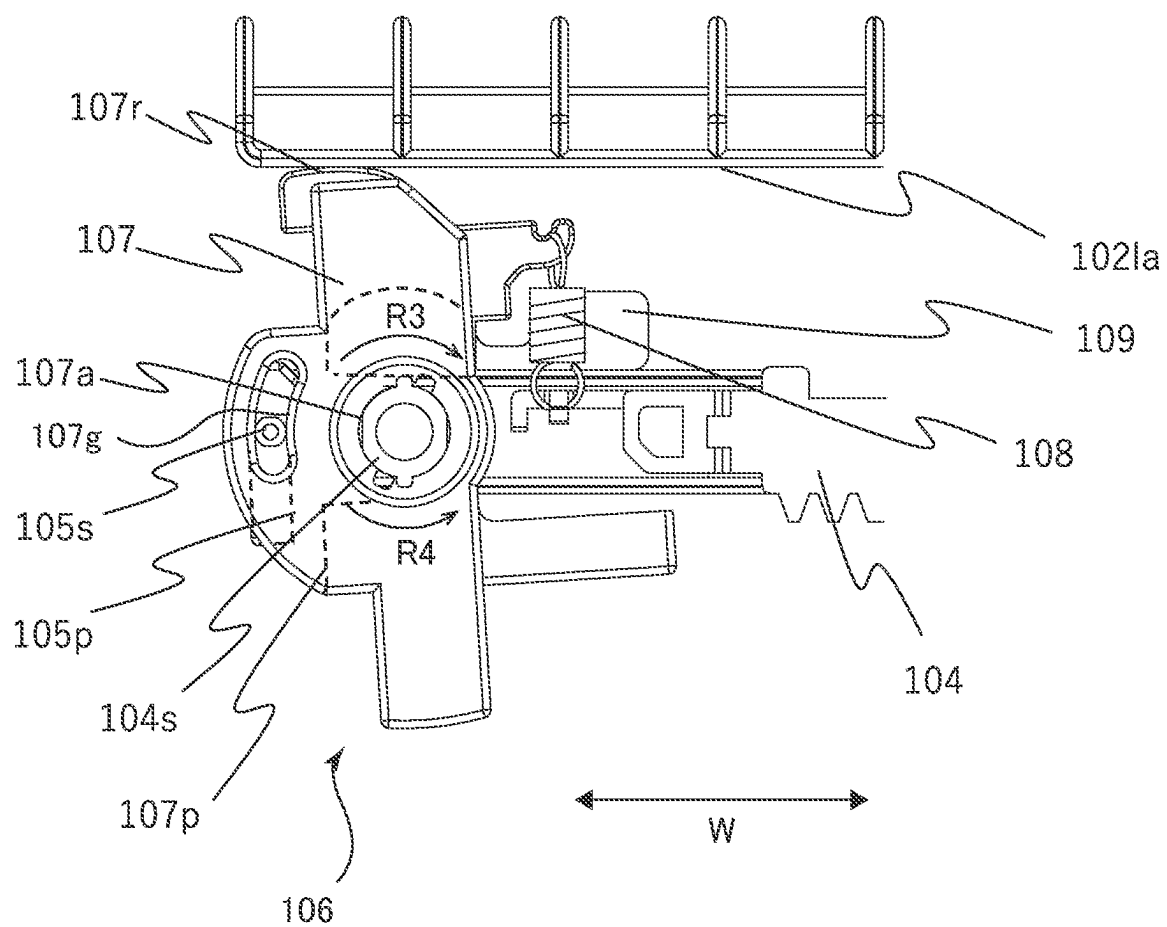


FIG. 15

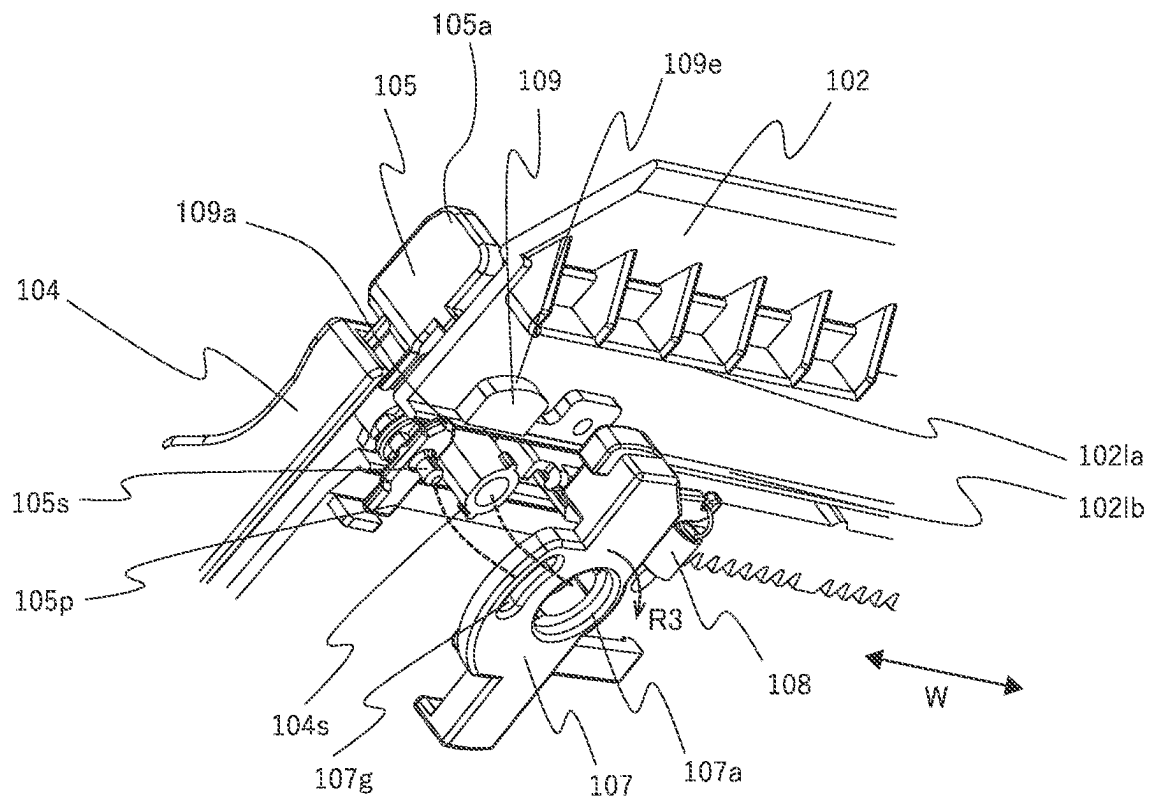


FIG.16

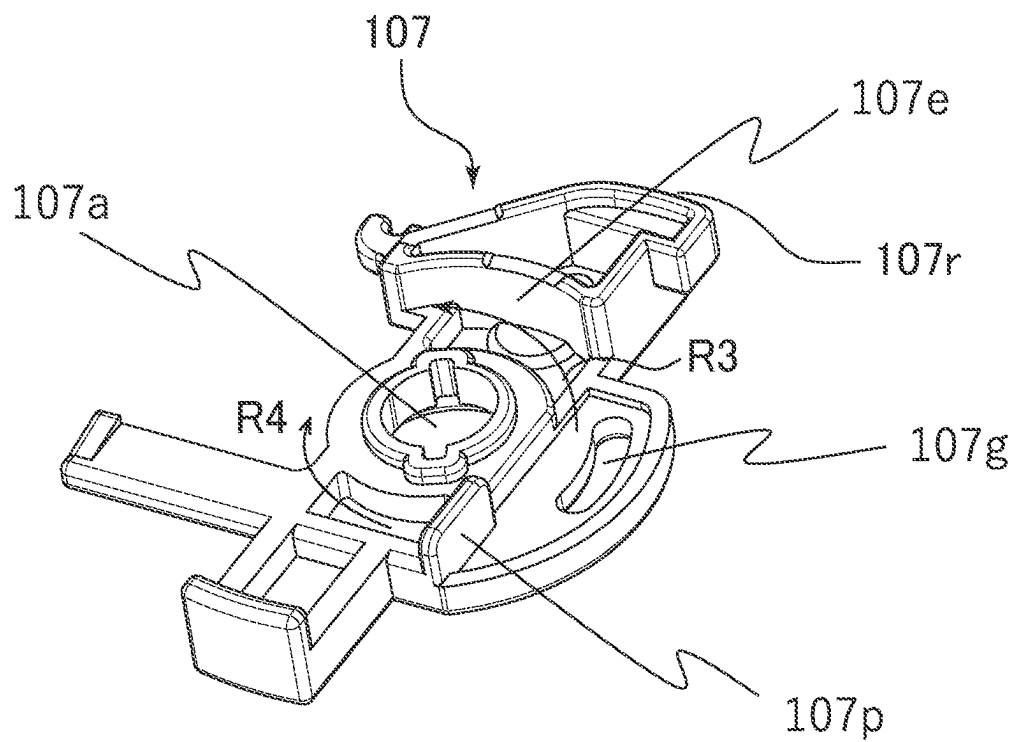


FIG. 17A

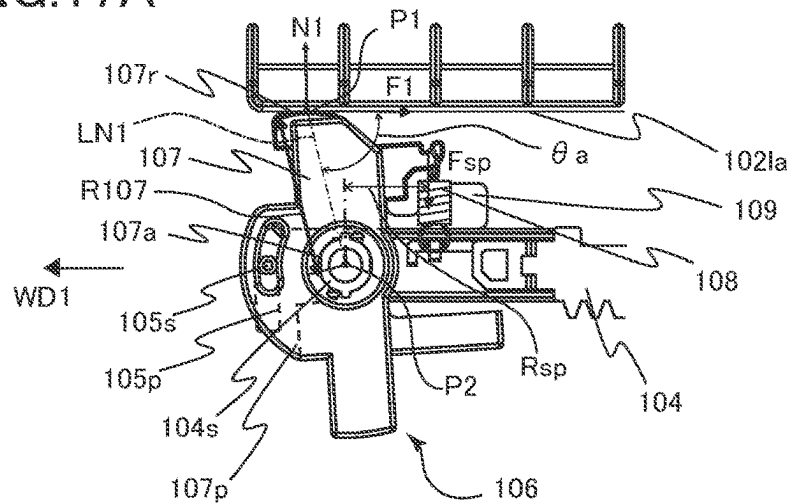


FIG. 17B

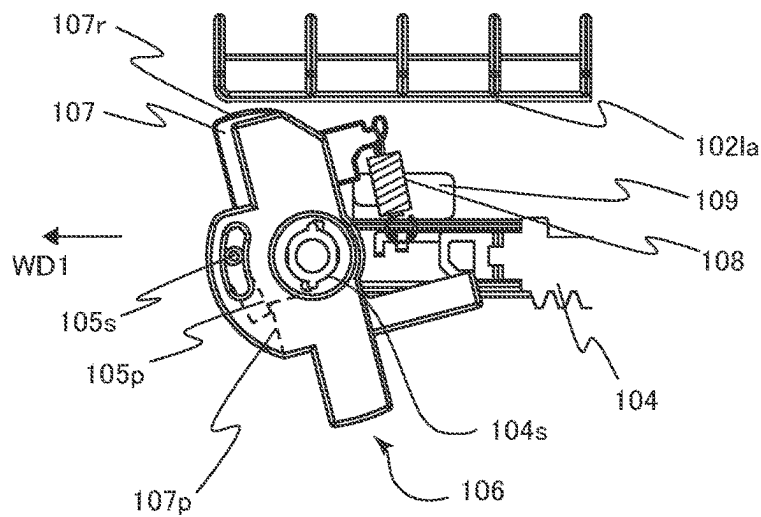
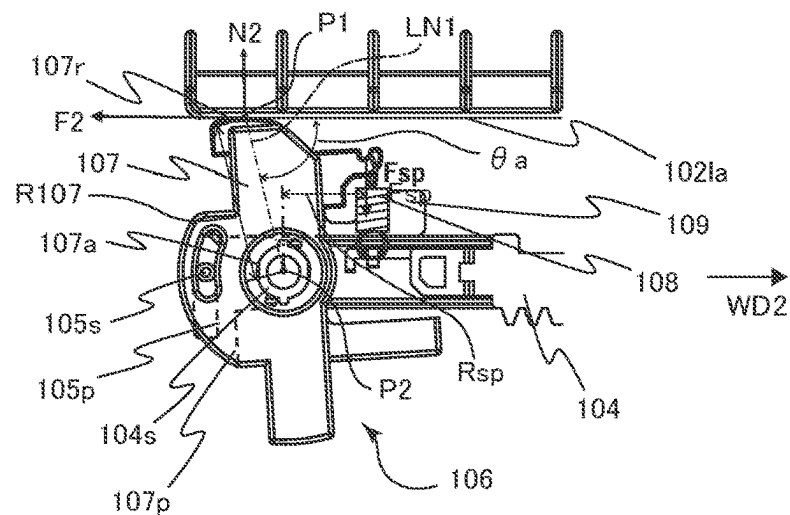


FIG. 17C



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SHEET CONVEYANCE APPARATUS, SHEET SUPPORT APPARATUS, AND IMAGE FORMING APPARATUS

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to a sheet conveyance apparatus that conveys a sheet, a sheet support apparatus that supports a sheet and an image forming apparatus.

Description of the Related Art

Japanese Patent Application Publication No. 2014-170058 proposes a copying machine that includes a housing and a double-sided cover. The double-sided cover is supported by the housing so as to be able to be opened and closed with respect to the housing by a user pivoting the double-sided cover on a cover pivot shaft. The copying machine is provided with a right-front handle portion and a right-back handle portion, which are used as handles when the copying machine is carried. The right-front handle portion and the right-back handle portion are disposed below the double-sided cover, and fixed to the housing.

However, since the right-front handle portion and the right-back handle portion are disposed below the double-sided cover, the cover pivot shaft of the double-sided cover described in Japanese Patent Application Publication No. 2014-170058 is positioned above the right-front handle portion and the right-back handle portion. Thus, the opening portion formed when the double-sided cover is opened becomes smaller, which causes insufficient maintainability.

In addition, the right-front handle portion and the right-back handle portion described in Japanese Patent Application Publication No. 2014-170058 are separate members, and disposed separated from each other. Thus, in the right side surface of the copying machine, more boundary lines between components are formed, which causes poor external appearance.

SUMMARY OF THE INVENTION

According to a first aspect of the present invention, a sheet conveyance apparatus includes an apparatus body including a conveyance portion configured to convey a sheet, and a door portion pivotally supported around a pivot axis disposed in a lower portion of the apparatus body such that the door portion is opened and closed with respect to the apparatus body. The apparatus body includes a holding member including a first holding portion used to lift the apparatus body, and the holding member is disposed inside a width of the door portion in an axis direction of the pivot axis. The holding member includes a first engaging portion and a second engaging portion disposed on the pivot axis. The door portion includes a first engaged portion configured to engage with the first engaging portion, and a second engaged portion configured to engage with the second engaging portion. The door portion is pivotally supported around the first engaging portion and the second engaging portion with respect to the holding member.

According to a second aspect of the present invention, a sheet support apparatus includes a sheet support portion configured to support a sheet, a regulation portion supported by the sheet support portion such that the regulation portion is configured to move in a first direction and a second direction opposite to the first direction, the regulation por-

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tion being configured to regulate a position of a downstream edge, in the first direction, of the sheet supported by the sheet support portion, and a brake unit configured to produce a first frictional force between the sheet support portion and the regulation portion in a case where the regulation portion moves toward the second direction, and produce a second frictional force larger than the first frictional force between the sheet support portion and the regulation portion in a case where the regulation portion moves toward the first direction.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an overall schematic diagram illustrating a printer of an embodiment.

FIG. 2A is a perspective view illustrating an image forming apparatus that is in a state where a conveyance cover is closed.

FIG. 2B is a perspective view illustrating the image forming apparatus that is in a state where the conveyance cover is opened.

FIG. 3 is an exploded perspective view illustrating the image forming apparatus that is in a state where the conveyance cover is detached.

FIG. 4A is a diagram illustrating a shaft-to-shaft distance of a comparative example.

FIG. 4B is a diagram illustrating a shaft-to-shaft distance of the present embodiment.

FIG. 5 is a perspective view illustrating an apparatus body that is in a state where the conveyance cover is detached.

FIG. 6 is a diagram of the conveyance cover viewed from the apparatus body side.

FIG. 7 is a perspective view illustrating a conveyance guide of the apparatus body.

FIG. 8 is a perspective view illustrating a locking member disposed on the conveyance cover.

FIG. 9A is a cross-sectional view illustrating a state where a handle is located at a first position and the locking member is located at a locking position.

FIG. 9B is a cross-sectional view illustrating a state where the handle is located at a second position and the locking member is located at an unlocking position.

FIG. 10A is a cross-sectional view illustrating a state where the conveyance cover is being opened while receiving an urging force of a spring.

FIG. 10B is a cross-sectional view illustrating a state where the locking member is being pushed downward by a hook portion.

FIG. 11 is a perspective view illustrating a manual-feed conveyance portion.

FIG. 12 is a perspective view illustrating side regulation plates.

FIG. 13 is a bottom perspective view illustrating the side regulation plates and a brake mechanism.

FIG. 14 is a front view illustrating the brake mechanism.

FIG. 15 is an exploded perspective view illustrating the brake mechanism.

FIG. 16 is a perspective view illustrating a brake member.

FIG. 17A is a diagram illustrating an operation of the brake mechanism performed when a side regulation plate is moved toward a first direction.

FIG. 17B is a diagram illustrating an operation of the brake mechanism that is in a state where a brake release lever is being operated.

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FIG. 17C is a diagram illustrating an operation of the brake mechanism performed when the side regulation plate is moved toward a second direction.

DESCRIPTION OF THE EMBODIMENTS

Overall Configuration

An image forming apparatus **1** of the present embodiment is an electrophotographic full-color laser-beam printer. As illustrated in FIG. 1, the image forming apparatus **1** includes an image forming portion **20** that forms an image on a sheet S, a sheet feeding portion **30**, a fixing apparatus **31**, and a manual-feed conveyance portion **2**. The image forming portion **20** includes four process cartridges **19Y**, **19M**, **19C**, and **19K**, and a scanner unit **21**. The four process cartridges **19Y**, **19M**, **19C**, and **19K** are used for forming four-color toner images of yellow (Y), magenta (M), cyan (C), and black (K), respectively.

Note that the four process cartridges **19Y**, **19M**, **19C**, and **19K** are the same in configuration as each other, except that they produce different colors of images. For this reason, the configuration and the image forming process of only the process cartridge **19Y** will be described, and the description for the process cartridges **19M**, **19C**, and **19K** will be omitted.

The process cartridge **19Y** includes a photosensitive drum **22**, a charging roller (not illustrated), and a developing roller **23**. The photosensitive drum **22** has an organic photoconductive layer coated on the outer surface of an aluminum cylinder, and is rotated by a driving motor (not illustrated). In addition, the image forming portion **20** includes an intermediate transfer belt **25**, and primary transfer rollers **24Y**, **24M**, **24C**, and **24K** disposed inside the intermediate transfer belt **25**.

The fixing apparatus **31** includes a fixing film that is heated by a heater (not illustrated), and a pressing roller that is in pressure contact with the fixing film. The sheet feeding portion **30** is disposed in a lower portion of the image forming apparatus **1**; and includes a cassette **11** that stores sheets, a pickup roller **12** that feeds a sheet, and a separation roller pair **13**.

Next, an image forming operation of the image forming apparatus **1** configured in this manner will be described. When the scanner unit **21** receives an image signal from an apparatus, such as a personal computer (not illustrated), the scanner unit **21** emits a laser beam, produced in accordance with the image signal, to the photosensitive drum **22** of the process cartridge **19Y**.

Since the surface of the photosensitive drum **22** is uniformly charged in advance by the charging roller (not illustrated) so as to have a predetermined polarity and potential, an electrostatic latent image is formed on the surface of the photosensitive drum **22** when the surface of the photosensitive drum **22** is irradiated with the laser beam from the scanner unit **21**. The electrostatic latent image formed on the photosensitive drum **22** is developed by the developing roller **23**, and a yellow (Y) toner image is formed on the photosensitive drum **22**.

Similarly, photosensitive drums of the process cartridges **19M**, **19C**, and **19K** are also irradiated with laser beams from the scanner unit **21**; and magenta (M), cyan (C), and black (K) toner images are formed on the photosensitive drums. The toner images formed on the photosensitive drums and having respective colors are transferred onto the intermediate transfer belt **25** by the primary transfer rollers **24Y**, **24M**, **24C**, and **24K**; and are conveyed to the secondary

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transfer roller **26** by the intermediate transfer belt **25** that rotates. Note that an image forming process for one color is performed at a timing at which a corresponding toner image is superposed on an upstream toner image which has been primary-transferred onto the intermediate transfer belt **25**.

In parallel with the image forming process, a sheet S is fed from the sheet feeding portion **30** or the manual-feed conveyance portion **2**. For example, a sheet S stored in the cassette **11** of the sheet feeding portion **30** is fed by the pickup roller **12**. The sheet S fed by the pickup roller **12** is separated from others, one by one, by the separation roller pair **13**; and is conveyed to a registration roller pair **15**.

The manual-feed conveyance portion **2** includes a manual feed cover **101**, a manual feed tray **102**, a pickup roller **16**, and a separation roller pair **17**. The manual feed cover **101** is supported by the conveyance cover **50** so as to be able to be opened and closed, and the manual feed tray **102** is held by the manual feed cover **101**. The sheet S stacked on the manual feed tray **102** is fed by the pickup roller **16**; then separated from others, one by one, by the separation roller pair **17**; and then conveyed to the registration roller pair **15**.

The registration roller pair **15** corrects the skew of the sheet S, and conveys the sheet S at a predetermined conveyance timing. Then, a full-color toner image on the intermediate transfer belt **25** is transferred onto the sheet S by a secondary transfer bias applied to the secondary transfer roller **26**.

The sheet S onto which the toner image has been transferred is then applied with predetermined heat and pressure by the fixing film and the pressing roller of the fixing apparatus **31**, so that the toner is melted and solidifies (fixed). Then, the sheet S having passed through the fixing apparatus **31** is discharged to a discharging tray **34** by a discharge roller pair **32**.

In a case where images are to be formed on both sides of the sheet S, after an image is fixed to a first side of the sheet S by the fixing apparatus **31**, the sheet S is conveyed toward a reversing roller pair **33**. Then the sheet S is switch-backed by the reversing roller pair **33**, and is conveyed and guided to a duplex conveyance path CP. The sheet S having been guided to the duplex conveyance path CP is conveyed again to the secondary transfer roller **26** by conveyance roller pairs **41** and **42** and the registration roller pair **15**. Then an image is formed on a second side of the sheet S by the secondary transfer roller **26**, and the image is fixed to the second side of the sheet S by the fixing apparatus **31**. After that, the sheet S is discharged to the discharging tray **34**.

The image forming apparatus **1** includes an apparatus body **1A** and the conveyance cover **50**. The apparatus body **1A** includes the image forming portion **20** and the sheet feeding portion **30**. The conveyance cover **50** serves as a door portion, and is supported so as to be able to be opened and closed with respect to the apparatus body **1A**. As described in detail below, the conveyance cover **50**, which is indicated by a dotted-line area of FIG. 1, is disposed on the right side surface of the image forming apparatus **1**; and the conveyance cover **50** supports a driven roller **14k** of the conveyance roller pair **14**, a driven roller of the conveyance roller pair **41**, a driven roller of the conveyance roller pair **42**, and the manual-feed conveyance portion **2**. The conveyance roller pair **14** includes a driving roller **14r** that serves as a first roller driven by a driving motor (not illustrated), and the driven roller **14k** that serves as a second roller rotated by the rotation of the driving roller **14r**. The driven roller **14k** is urged by a spring **43**, which serves as an urging member, so that the driven roller **14k** is in pressure contact with the driving roller **14r**. Note that the driven roller **14k**

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may serve as a driving roller and the driving roller **14r** may serve as a driven roller. In another case, another elastic member, such as a rubber member or a sponge member, may be used in place of the spring **43**.

Peripheral Configuration of Conveyance Cover

Next, a peripheral configuration of the conveyance cover **50** will be described with reference to FIGS. **2A** to **3**. As illustrated in FIG. **2A**, the conveyance cover **50** is disposed on the right side surface of the image forming apparatus **1**, and the manual feed cover **101** is supported by the conveyance cover **50** so as to be able to be opened and closed. The conveyance cover **50** can be opened by a user operating a handle **51**. The manual feed cover **101** includes a recess portion **101a**, which is disposed below the handle **51**; and can be opened and closed with respect to the conveyance cover **50**, by a user operating the recess portion **101a**.

As illustrated in FIGS. **2A** and **2B**, the apparatus body **1A** includes a frame **80** and a handle **70**. The handle **70** serves as a holding member fixed to the frame **80**. Since the handle **70** is fixed to the frame **80** as illustrated in FIG. **2B**, the position and posture of the handle **70** do not change even when the conveyance cover **50** is opened and closed. As illustrated in FIG. **2A**, the handle **70** includes a holding portion **70h**, which serves as a first holding portion used to lift the apparatus body **1A** by a user. The holding portion **70h** is recessed with respect to an exterior surface **50d** of the conveyance cover **50** in a state where the conveyance cover **50** is closed with respect to the apparatus body **1A**.

In addition, the handle **70** is disposed in a lower portion of the apparatus body **1A** and inside the width of the conveyance cover **50** in a width direction **W**. In other words, the front edge of the handle **70** is positioned behind the front edge of the conveyance cover **50**, and the back edge of the handle **70** is positioned in front of the back edge of the conveyance cover **50**.

As illustrated in FIG. **3**, the handle **70** includes a first shaft portion **70a** and a second shaft portion **70b**. The first shaft portion **70a** serves as a first engaging portion that is disposed on a lower edge portion of a first end surface **70c**. The first end surface **70c** is the surface of one end portion of the handle **70** in the width direction **W**. The second shaft portion **70b** serves as a second engaging portion that is disposed on a lower edge portion of a second end surface **70d**. The second end surface **70d** is the surface of the other end portion of the handle **70** in the width direction **W**. That is, the holding portion **70h** is disposed between the first shaft portion **70a** and the second shaft portion **70b** in the width direction **W**. The first shaft portion **70a** and the second shaft portion **70b** are shaft portions that extend in directions, one of which extends away from the other in the width direction **W**.

In addition, the conveyance cover **50** includes a first hole portion **50a** and a second hole portion **50b**. The first hole portion **50a** serves as a first engaged portion that engages with the first shaft portion **70a**. The second hole portion **50b** serves as a second engaged portion that engages with the second shaft portion **70b**. The conveyance cover **50** is pivotally supported around the first shaft portion **70a** and the second shaft portion **70b** with respect to the handle **70**. In other words, the first shaft portion **70a**, the second shaft portion **70b**, the first hole portion **50a**, and the second hole portion **50b** are disposed on a pivot axis **50R** that is a center of pivot of the conveyance cover **50**. The axis direction of the pivot axis **50R** is parallel with the width direction **W**. That is, the conveyance cover **50** is pivotally supported by the apparatus body **1A** around the pivot axis **50R** disposed in a lower portion of the apparatus body **1A**.

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Next, an image forming apparatus **1000** will be described as a comparative example with reference to FIG. **4A**. As illustrated in FIG. **4A**, the image forming apparatus **1000** has two handles **170** and **270**. The handle **170** is disposed in a front edge portion of the image forming apparatus **1000**, and the handle **270** is disposed in a back edge portion of the image forming apparatus **1000**. A conveyance cover **150** of the image forming apparatus **1000** is pivotally supported by a first shaft portion **170a** disposed on the handle **170**, and by a second shaft portion **170b** disposed on the handle **270**.

Each of the handles **170** and **270** has a width **W1**, which is necessary for a user to hold the handle for carrying the image forming apparatus **1000**. Thus, a shaft-to-shaft distance **L1** between the first shaft portion **170a** and the second shaft portion **170b** in the width direction **W** is shortened by double the width **W1** of each of the two handles **170** and **270**.

If the shaft-to-shaft distance **L1** is shortened, the misalignment of the conveyance cover **150** produced in the attachment of the conveyance cover **150** will have more affect on the inclination of the conveyance cover **150** with respect to the pivot center axis. In addition, when the conveyance cover **150** is closed, the position of the conveyance cover **150** may deviate from a nominal position. Since the conveyance cover **150** supports the driven rollers of the conveyance roller pairs **14**, **41**, and **42**, the stability of conveyance of sheets will deteriorate if the position of the conveyance cover **150** deviates from the nominal position.

In contrast, in the present embodiment, only one handle **70** is disposed in the substantially central portion of the conveyance cover **50** in the width direction **W**, as illustrated in FIGS. **4B** and **5**. FIG. **5** is a perspective view illustrating the apparatus body **1A** that is in a state where the conveyance cover **50** is detached from the apparatus body **1A**. As can be seen, the holding portion **70h** of the handle **70** has a width **W2** that is sufficient for a user to hold the holding portion **70h** for carrying the apparatus body **1A** of the image forming apparatus **1**.

As described above, the first shaft portion **70a** and the second shaft portion **70b**, which pivotally support the conveyance cover **50**, are respectively disposed on the first end surface **70c** and the second end surface **70d** of the handle **70**. Thus, even in a state where the handle **70** has the width **W2**, it is possible to secure a shaft-to-shaft distance **L2** between the first shaft portion **70a** and the second shaft portion **70b** in the width direction **W**. The shaft-to-shaft distance **L2** is larger than the shaft-to-shaft distance **L1**.

FIG. **6** is a front view of the conveyance cover **50** closed with respect to the apparatus body **1** and viewed from the apparatus body **1A** side. As illustrated in FIG. **6**, the conveyance cover **50** has a width **L50** in the width direction **W**. In addition, the conveyance cover **50** has a conveyance guide **52**, which constitutes a part of the duplex conveyance path **CP** (see FIG. **1**). The duplex conveyance path **CP** is a conveyance path through which the sheet **S**, conveyed by the reversing roller pair **33** that serves as a conveyance portion, passes. The conveyance guide **52** has a width **Lp**, which is the length of the conveyance guide **52** in the width direction **W**. Note that the conveyance guide **52** may be formed integrated with the conveyance cover **50**, or may be a separate member different from the conveyance cover **50**.

The above-described shaft-to-shaft distance **L2** of the present embodiment is set equal to or larger than two thirds of the width **L50** of the conveyance cover **50**. In addition, the shaft-to-shaft distance **L2** is larger than the width **Lp** of the conveyance guide **52**. Since the shaft-to-shaft distance **L2** is set longer as much as possible, the misalignment of the

conveyance cover 50 produced in the attachment of the conveyance cover 50 have less affect on the inclination of the conveyance cover 50 with respect to the pivot axis 50R. As a result, it becomes possible to improve the operational feeling that a user feels when opening and closing the conveyance cover 50. In addition, it becomes possible to suppress the deviation of the position of the conveyance cover 50 from a nominal position, and improve the stability of conveyance of sheets.

In addition, in the present embodiment, the first shaft portion 70a is disposed on a lower edge portion of the first end surface 70c of the handle 70, and the second shaft portion 70b is disposed on a lower edge portion of the second end surface 70d. With this arrangement, it is possible to dispose the pivot axis 50R of the conveyance cover 50 in a lower edge portion of the handle 70 and in a lower edge portion of the image forming apparatus 1. Thus, the opening formed when the conveyance cover 50 is opened can be made larger. As a result, it becomes possible to improve the maintainability, which is necessary for a user to perform the jam handling, the replacement of a process cartridge, the maintenance of the interior of the apparatus body 1A, and the like.

Furthermore, in the present embodiment, since the handle 70 is not divided into two members and is a single member, boundary lines between components can be made simple in the right side surface of the image forming apparatus 1. More specifically, as illustrated in FIG. 4B, since a boundary line 70L between the handle 70 and the conveyance cover 50 is a single continuous line, the external appearance of the image forming apparatus 1 can be improved. In addition, since the number of components is made smaller than that in the configuration in which two handles are disposed, the cost can be reduced.

Lock Mechanism

Next, a lock mechanism will be described. The lock mechanism causes the apparatus body 1A to hold the conveyance cover 50 in a state where the conveyance cover 50 is closed. As illustrated in FIG. 7, a conveyance guide 81 is fixed to the frame 80 of the apparatus body 1A. The conveyance guide 81 constitutes a part of the duplex conveyance path CP (see FIG. 1), and guides the sheet S. The conveyance guide 81 serves as a locked member, and includes hook portions 81a and 81b.

In addition, as illustrated in FIG. 8, the conveyance guide 52 is fixed to the conveyance cover 50. The conveyance guide 52 constitutes a part of the duplex conveyance path CP (see FIG. 1), and supports a locking member 53 such that the locking member 53 can move. The conveyance guide 52 includes a guide surface 52a that guides the sheet S. The locking member 53 includes locking hole portions 53a and 53b that can respectively engage with the hook portions 81a and 81b disposed in the conveyance guide 81 of the apparatus body 1A.

More specifically, the locking member 53 can move between a locking position (see FIG. 9A) and an unlocking position (see FIG. 9B), with respect to the conveyance guide 52. When the locking member 53 is located at the locking position, the locking hole portions 53a and 53b engage with the hook portions 81a and 81b, and the locking member 53 locks the conveyance cover 50 on the apparatus body 1A. When the locking member 53 is located at the unlocking position, the locking hole portions 53a and 53b are separated from the hook portions 81a and 81b, and the locking member 53 allows the conveyance cover 50 to be opened and closed with respect to the apparatus body 1A.

Next, with reference to FIG. 2 and FIGS. 9A and 9B, a more detailed configuration of the lock mechanism will be described. As illustrated in FIGS. 2A and 9A, the conveyance cover 50 supports the handle 51, which serves as a cover member, such that the handle 51 can pivot on an opening-and-closing axis 51c and can be opened and closed. More specifically, the handle 51 has an exterior surface 51a that constitutes a part of the exterior of the image forming apparatus 1. The handle 51 can move between a first position illustrated in FIG. 9A, and a second position illustrated in FIG. 9B. When the handle 51 is located at the first position, the exterior surface 51a is flush with the exterior surface 50d of the conveyance cover 50. If a user presses the exterior surface 51a, the handle 51 is pushed inward, and moved from the first position to the second position. At the first position, the handle 51 covers a below-described holding portion 50h.

The handle 51 is urged, clockwise in FIG. 9A, by a spring 55 toward the first position. An abutment surface 51b formed on the handle 51 abuts against the conveyance cover 50, so that the handle 51 is positioned at the first position.

The conveyance guide 52 of the conveyance cover 50 is provided with a slide guide 52S that extends in a vertical direction VD. The locking member 53 includes a slide rib 53R that can engage with the slide guide 52S. The locking member 53 is supported such that the engagement between the slide guide 52S and the slide rib 53R allows the locking member 53 to slide with respect to the conveyance guide 52 in the vertical direction VD. The locking member 53 is urged by a spring 54, which serves as an urging member, toward the above-described locking position, and is positioned at the locking position by a stopper (not illustrated).

Next, with reference to FIGS. 9A to 10B, the operation of the handle 51, the locking member 53, and the conveyance cover 50 will be described. As illustrated in FIG. 9A, in a state where the conveyance cover 50 is closed with respect to the apparatus body 1A, the handle 51 is positioned at the first position, and the locking member 53 is positioned at the locking position.

As illustrated in FIGS. 9A and 9B, when a user opens the conveyance cover 50 with respect to the apparatus body 1A, the user presses the exterior surface 51a of the handle 51 with a finger. With this operation, the handle 51 pivots counterclockwise on the opening-and-closing axis 51c against the urging force of the spring 55, and moves from the first position to the second position. After that, the user can access the holding portion 50h disposed above the handle 51 and inside the conveyance cover 50. Thus, the user can hook a finger to the holding portion 50h, which serves as a second holding portion, and hold the holding portion 50h.

While the handle 51 moves from the first position to the second position, the handle 51 presses the locking member 53 downward against the urging force of the spring 54. With this operation, the locking member 53 slides to the unlocking position, as illustrated in FIG. 9B. Since the locking member 53 is pushed downward from the locking position to the unlocking position, the hook portions, 81a and 81b, and the locking hole portions, 53a and 53b, are disengaged from each other, and the conveyance cover 50 is allowed to be opened and closed with respect to the apparatus body 1A.

As described with reference to FIG. 1, the conveyance cover 50 rotatably supports the driven roller 14k, and in a state where the conveyance cover 50 is closed, the driven roller 14k is urged by the spring 43 so as to be in pressure contact with the driving roller 14r. In addition, the driven roller 14k receives a reaction force, due to an urging force of

the spring 43, from the driving roller 14r in a direction in which the conveyance cover 50 is opened.

As illustrated in FIG. 9B, when the locking member 53 moves to the unlocking position, the conveyance cover 50 is moved with respect to the apparatus body 1A and opened, by the above-described reaction force. With this operation, the conveyance cover 50 is moved to the open position and opened, as illustrated in FIG. 2B. Note that a user may move the conveyance cover 50 to the open position for opening the conveyance cover 50, while holding the holding portion 50h, after pressing and moving the handle 51 from the first position to the second position.

When a user closes the conveyance cover 50 with respect to the apparatus body 1A, the user holds one portion of the conveyance cover 50 and pivots the conveyance cover 50 toward the closing direction, on the pivot axis 50R (see FIG. 2B), as illustrated in FIG. 10B. As a result, the locking member 53 abuts against sloped surfaces 81c and 81d of the hook portions 81a and 81b. The sloped surfaces 81c and 81d are sloped such that as the sloped surfaces 81c and 81d extend closer to the photosensitive drum 22 (see FIG. 1) in the horizontal direction, the sloped surfaces 81c and 81d extend more downward in the vertical direction VD.

If the conveyance cover 50 is further closed in this state, the locking member 53 is pressed downward by the sloped surfaces 81c and 81d against the urging force of the spring 54. If the edge portion of the locking member 53 climbs over the sloped surfaces 81c and 81d, and the hook portions, 81a and 81b, and the locking hole portions, 53a and 53b, engage with each other, the locking member 53 is pushed upward to the locking position by the urging force of the spring 54. With this operation, the conveyance cover 50 is locked in a state where the conveyance cover 50 is closed with respect to the apparatus body 1A.

Since the lock mechanism is configured in this manner, the handle 51 is flush with the exterior surface 50d of the conveyance cover 50 when located at the first position and applied with no external force. Thus, the right side surface of the image forming apparatus 1 has no projecting and recess portions to open the conveyance cover 50, so that the external appearance can be improved.

In addition, as illustrated in FIG. 9A, the locking member 53 is disposed in a space SP formed between the guide surface 52a of the conveyance guide 52 and the exterior surface 50d of the conveyance cover 50 in the horizontal direction. Since the image forming apparatus 1 is required to be downsized, the space SP needs to be made smaller. For this reason, in the present embodiment, the locking member 53 is configured to slide and move in the vertical direction VD. Thus, the locking member 53 can be disposed in the small space SP, and the amount of engagement between the locking hole portions, 53a and 53b, and the hook portions, 81a and 81b, in the vertical direction VD can be maximized. That is, the downsizing of the lock mechanism and the image forming apparatus 1, and the reliable lock of the conveyance cover 50 to the apparatus body 1A can be both achieved.

Furthermore, in the present embodiment, when a user moves the handle 51 from the first position to the second position, the locking member 53 moves to the unlocking position in accordance with the movement of the handle 51, and the conveyance cover 50 is automatically opened by the urging force of the spring 43. Thus, the need for troublesome operation for opening the conveyance cover 50 can be eliminated, so that the usability can be improved. In addition, since a user can intuitively hold the holding portion 50h that appears simultaneously when the user pushes and opens

the handle 51, the user can open the conveyance cover 50 while holding the holding portion 50h. As a result, the shock caused when a user opens the conveyance cover 50 can be reduced.

5 Manual-Feed Conveyance Portion

Next, a lock configuration of side regulation plates 103 and 104 of the manual-feed conveyance portion 2 will be described. As illustrated in FIGS. 1 and 11, the manual-feed conveyance portion 2, which serves as a sheet support apparatus, includes a pickup arm 16a and a pickup roller 16. The pickup roller 16 is supported by the pickup arm 16a so as to be able to move up and down. In addition, the manual-feed conveyance portion 2 includes a manual feed cover 101, a manual feed tray 102, tray links 121 and 122, and the side regulation plates 103 and 104. The manual feed tray 102 serves as a sheet support portion. The pickup roller 16 feeds the sheet S stacked on the manual feed tray 102, in a feeding direction FD.

The manual feed cover 101 is supported so as to be opened and closed with respect to the conveyance cover 50, and constitutes a part of the exterior surface of the apparatus. On the top of the manual feed cover 101, the manual feed tray 102 is held. The manual feed cover 101 can be positioned at a predetermined degree of opening by the tray links 121 and 122. One end portion of each of the tray links 121 and 122 is linked to the manual feed tray 102, and the manual feed tray 102 can move in accordance with the opening/closing operation of the manual feed cover 101 along the manual feed cover 101. In this configuration, when the manual feed cover 101 is closed, the manual feed tray 102 is retracted from the pickup arm 16a and the pickup roller 16. Thus, the manual-feed conveyance portion 2 can be downsized.

As illustrated in FIG. 12, in the manual feed tray 102, guide grooves Lm102 and Ln102 that extend in the width direction W are formed. The side regulation plate 103 is held by the manual feed tray 102 so as to be able to move in the width direction W, along the guide groove Lm102. Similarly, the side regulation plate 104, which serves as a regulation portion, is held by the manual feed tray 102 so as to be able to move in the width direction W, along the guide groove Ln102.

As illustrated in FIG. 13, the side regulation plates 103 and 104 are respectively provided with rack-and-gear portions 103R and 104R, which extend in the width direction W. In addition, a pinion gear 200 that meshes with the rack-and-gear portions 103R and 104R is rotatably supported by the bottom portion of the manual feed tray 102. Thus, the side regulation plates 103 and 104 are linked with each other via the rack-and-gear portions 103R and 104R and the pinion gear 200, and move together so as to move in directions opposite to each other in the width direction W.

One direction in the width direction W is defined as a first direction WD1, and a direction opposite to the first direction WD1 is defined as a second direction WD2. That is, the first direction WD1 and the second direction WD2 are orthogonal to the feeding direction FD. For example, the side regulation plates 103 and 104 move together such that if the side regulation plate 103 is moved by a user toward the first direction WD1, the side regulation plate 104 moves toward the second direction WD2. The side regulation plate 103 includes a regulation surface 103a that regulates the position of a downstream edge of the sheet S stacked on the manual feed tray 102. The downstream edge is an edge of the sheet S, located downstream in the second direction WD2. The side regulation plate 104 includes a regulation surface 104a that regulates the position of a downstream edge of the sheet

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S stacked on the manual feed tray 102. The downstream edge is an edge of the sheet S, located downstream in the first direction WD1.

For example, by a user operating the side regulation plate 104, both edge portions of the sheet S (stacked on the manual feed tray 102), located in the width direction W, are nipped by the regulation surfaces 103a and 104a, so that the position of the sheet S in the width direction W is regulated. The manual-feed conveyance portion 2 is provided with a brake mechanism 106, which serves as a brake unit that generates frictional force (brake force) for preventing the movement of the side regulation plate 104.

Brake Mechanism

Next, a configuration of the brake mechanism 106 will be described with reference to FIGS. 14 to 16. As illustrated in FIGS. 14 to 16, the brake mechanism 106 includes a brake member 107, a spring 108, a brake wall 102a, a rail wall 102b, a brake piece 109, and a brake release lever 105.

The side regulation plate 104 includes a shaft portion 104s that engages with a hole portion 107a of the brake member 107. Thus, the brake member 107 is supported so as to be able to pivot on the shaft portion 104s of the side regulation plate 104. The brake wall 102a and the rail wall 102b are disposed on the manual feed tray 102, and extend in the width direction W. Note that the brake wall 102a and the rail wall 102b are disposed, shifted from each other in the sheet conveyance direction.

The brake member 107 includes a brake surface 107r that can abut against the brake wall 102a, a circumferential surface 107e that is formed around the hole portion 107a, a long hole 107g that is formed around the hole portion 107a and extending like an arc, and a release surface 107p. The brake member 107 is urged by an urging force of the spring 108, which serves as a brake urging portion, toward a direction indicated by an arrow R3 around the shaft portion 104s; and the brake surface 107r is pressed against the brake wall 102a.

As illustrated in FIG. 15, the brake piece 109 is sandwiched between the manual feed tray 102 and the brake member 107 in the thickness direction of the brake piece 109. The brake piece 109 includes a circumferential surface 109e that engages with the circumferential surface 107e of the brake member 107, and a slide surface 109a that can slide on the rail wall 102b.

The spring 108 produces the contact force applied between the brake surface 107r and the brake wall 102a, and the reaction force of the contact force produces the nipping force that causes the circumferential surface 107e of the brake member 107 and the rail wall 102b to nip the brake piece 109. In other words, the brake member 107 and the brake piece 109 are nipped between the brake wall 102a and the rail wall 102b.

The brake member 107, the spring 108, the brake piece 109, and the brake release lever 105 are arranged so as to move together with the side regulation plate 104, in the width direction W. However, if the brake piece 109 is nipped by the circumferential surface 107e of the brake member 107 and the rail wall 102b, the brake piece 109 applies brake force so that the side regulation plate 104 does not move in the width direction W.

Note that since the brake wall 102a and the rail wall 102b extend in the width direction W, the brake mechanism 106 can apply the brake force to the side regulation plate 104, regardless of the position of the side regulation plate 104 in the width direction W.

As illustrated in FIGS. 14 to 16, the brake release lever 105, which serves as a brake release member, is pivotally

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supported by the side regulation plate 104. The brake release lever 105 includes a shaft 105s, a holding portion 105a, and a release portion 105p. The shaft 105s is disposed on the pivot axis of the brake release lever 105, and loosely fits in the long hole 107g of the brake member 107.

For releasing the brake of the side regulation plate 104, a user pivots the brake release lever 105 on the shaft 105s, toward a direction indicated by an arrow R1 (see FIG. 12), by operating the holding portion 105a. As a result, the release portion 105p abuts against the release surface 107p of the brake member 107. If the brake release lever 105 is further pivoted, in this state, toward the direction indicated by the arrow R1, the brake member 107 pivots on the shaft portion 104s, toward a below-described brake release position (see FIG. 17B). When the brake release lever 105 is positioned at the brake release position, the brake of the side regulation plate 104 applied by the brake mechanism 106 is released.

Operation of Brake Mechanism

Next, an operation of the brake mechanism 106 will be described more specifically with reference to FIGS. 17A to 17C. As described above, the brake mechanism 106 of the present embodiment can release the brake force applied to the side regulation plate 104, by a user operating the brake release lever 105. In addition, in a state where the brake release lever 105 is not operated by a user, the operation of the brake mechanism 106 performed when the external force is applied to the side regulation plate 104 in the first direction WD1 is different from the operation of the brake mechanism 106 performed when the external force is applied to the side regulation plate 104 in the second direction WD2. Hereinafter, the operation of the brake mechanism 106 performed when the external force is applied to the side regulation plate 104 in the first direction WD1, and the operation of the brake mechanism 106 performed when the external force is applied to the side regulation plate 104 in the second direction WD2 will be described sequentially.

First, with reference to FIG. 17A, the operation of the brake mechanism 106 performed when the external force is applied to the side regulation plate 104 in the first direction WD1, in a state where the brake release lever 105 is not operated by a user, will be described. As one example, the case where the external force is applied to the side regulation plate 104 in the first direction WD1 is a case where the regulation surface 104a of the side regulation plate 104 receives force from the sheet S when the sheet S stacked on the manual feed tray 102 is fed by the pickup roller 16. As another example, the case where the external force is applied to the side regulation plate 104 in the first direction WD1 is a case where a user tries to manually move the side regulation plate 104 in the first direction WD1.

As illustrated in FIG. 17A, if the external force tries to move the side regulation plate 104 toward the first direction WD1, a frictional force F1 that serves as a second frictional force is produced. The frictional force F1 is brake force produced between the brake surface 107r of the brake member 107 and the brake wall 102a. When the frictional force F1 is produced, the brake surface 107r exerts a normal force N1 on the brake wall 102a. The frictional force F1 and the normal force N1 are expressed by the following computational equation.

$$F1 = \mu \times N1 \quad (1)$$

$$N1 = F1 \times \tan \theta a + (Fsp \times Rsp) / (R107 \times \cos \theta a) \quad (2)$$

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μ : a coefficient of friction between the brake surface **107r** and the brake wall **102/a**

θa : an angle between a straight line LN1 and the brake wall **102/a**, where the straight line LN1 passes through a contact point P1 between the brake surface **107r** and the brake wall **102/a** and a center P2 of the shaft portion **104s**

R107: a distance between the contact point P1 and the center P2

Fsp: a force produced by the spring **108**

Rsp: a radius in the rotation moment that the spring **108** gives the brake member **107**

From the above-described equations (1) and (2), the frictional force F1 that is the brake force is expressed by the following equation (3).

$$F1 = [\mu \times (Fsp \times Rsp) / (R107 \times \cos \theta a)] / (1 - \mu \times \tan \theta a) \quad (3)$$

From equation (3), it is understood that even when the coefficient of friction and the force Fsp produced by the spring **108** have small values, the frictional force F1 has a relatively large value if the angle θa has a value near 90 degrees. For example, if μ is 0.3, R107 is 20 mm, Fsp is 150 gf, Rsp is 20 mm, and θa is 70 degrees, the frictional force F1 is 748.5 gf. Thus, even if the external force is applied to the side regulation plate **104** in the first direction WD1, the frictional force F1 can sufficiently hold the side regulation plate **104** so that the side regulation plate **104** does not move.

As is clear from the above-described equations (1) and (2), the frictional force F1 includes a force, $F1 \times \tan \theta a$, which is applied as the brake surface **107r** bites into the brake wall **102/a**. For example, if a user tries to move the side regulation plate **104** toward the first direction WD1 while the contact point P1 hardly moves due to the frictional force F1, the angle θa increases. Note that the angle θa is larger than 0 degrees and smaller than 90 degrees. If a user tries to move the side regulation plate **104** toward the first direction WD1, the force, $F1 \times \tan \theta a$, increases, increasing the frictional force F1. In other words, the frictional force F1 increases as the side regulation plate **104** moves more downstream in the first direction WD1 in a state where the brake surface **107r** is in contact with the brake wall **102/a**.

Thus, even if the sheet S presses the regulation surface **104a** of the side regulation plate **104** for example, the position of the side regulation plate **104** is kept by the brake mechanism **106**, so that the failure in conveyance of sheets, such as skew of sheets, can be reduced.

Next, with reference to FIG. 17B, the operation of the brake mechanism **106** performed when a user tries to move the side regulation plate **104** toward the first direction WD1 while pivoting the brake release lever **105** toward the direction indicated by the arrow R1 will be described. If a user operates and pivots the brake release lever **105** toward the direction indicated by the arrow R1 (see FIG. 12), the brake member **107** pivots and moves to the brake release position, as illustrated in FIG. 17B.

As a result, the brake surface **107r** of the brake member **107** is separated from the brake wall **102/a** of the manual feed tray **102**, and the above-described frictional force F1 disappears. Thus, the user can move the side regulation plate **104** toward the first direction WD1, with weak force in a state where the frictional force F1 is not produced. In addition, even when the user tries to move the side regulation plate **104** toward the second direction WD2 while pivoting the brake release lever **105** toward the direction indicated by the arrow R1, the frictional force F1 is not produced.

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Next, with reference to FIG. 17C, the operation of the brake mechanism **106** performed when the external force is applied to the side regulation plate **104** in the second direction WD2, in a state where the brake release lever **105** is not operated by a user, will be described. For example, the case where the external force is applied to the side regulation plate **104** in the second direction WD2 is a case where a user tries to manually move the side regulation plate **104** toward the second direction WD2.

As illustrated in FIG. 17C, if the external force tries to move the side regulation plate **104** toward the second direction WD2, a frictional force F2 that serves as a first frictional force is produced. The frictional force F2 is brake force produced between the brake surface **107r** and the brake wall **102/a**. When the frictional force F2 is produced, the brake surface **107r** exerts a normal force N2 on the brake wall **102/a**. The frictional force F2 and the normal force N2 are expressed by the following computational equation.

$$F2 = \mu \times N2 \quad (4)$$

$$N2 = (Fsp \times Rsp) / (R107 \times \cos \theta a) \quad (5)$$

From the above-described equations (4) and (5), the frictional force F2 that is the brake force is expressed by the following equation (6).

$$F2 = \mu \times (Fsp \times Rsp) / (R107 \times \cos \theta a) \quad (6)$$

If a user tries to move the side regulation plate **104** toward the second direction WD2, the brake surface **107r** behaves so as to separate from the brake wall **102/a**, due to the frictional force F2. Thus, as is clear from the above-described equation (6) and is different from the description made with reference to FIG. 17A, the frictional force F2 does not include the force, $F1 \times \tan \theta a$, which is applied as the brake surface **107r** bites into the brake wall **102/a**.

Similar to the parameters described with reference to FIG. 17A, if μ is 0.3, R107 is 20 mm, Fsp is 150 gf, Rsp is 20 mm, and θa is 70 degrees, the frictional force F2 is 131.5 gf. That is, the above-described frictional force F1 is larger than the frictional force F2. In addition, since the frictional force F2 does not include the above-described force, $F1 \times \tan \theta a$, the brake surface **107r** slides on the brake wall **102/a** and the angle θa is unchanged even if the side regulation plate **104** is moved toward the second direction WD2. In other words, even if the side regulation plate **104** is moved downstream in the second direction WD2 in a state where the brake surface **107r** is in contact with the brake wall **102/a**, the frictional force F2 is constant.

Thus, a user can move the side regulation plate **104** toward the second direction WD2 against the frictional force F2 with relatively weak force, without moving the brake release lever **105** toward the direction indicated by the arrow R1. That is, a user can easily move the side regulation plates **103** and **104** toward a direction in which the side regulation plates **103** and **104** move closer to the sheet S stacked on the manual feed tray **102**. In this manner, a user can easily regulate the position of the sheet S in the width direction W. Thus, the usability can be improved.

In addition, the brake mechanism **106** of the present embodiment can steplessly regulate the position of the side regulation plate **104** in the width direction W. Thus, the distance between an edge portion of the sheet S in the width direction W and the regulation surface **103a** of the side regulation plate **103**, and the distance between an edge portion of the sheet S in the width direction and the regulation surface **104a** of the side regulation plate **104** can be made smaller, so that the sheet S can be positioned with

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high accuracy. For example, there is a case in which the size of the sheet S differs from the nominal size of the sheet S due to a tolerance produced in the cutting process of the sheet S. Even in such a case, since the distance between an edge portion of the sheet S in the width direction W and the regulation surface **103a**, and the distance between an edge portion of the sheet S in the width direction and the regulation surface **104a** can be minimized, the failure in conveyance of sheets, such as skew of sheets, can be reduced.

Other Embodiments

In the present embodiment, the first shaft portion **70a** and the second shaft portion **70b** are disposed in the handle **70**, and the first hole portion **50a** and the second hole portion **50b** are disposed in the conveyance cover **50**. However, the present disclosure is not limited to this. For example, the first hole portion and the second hole portion may be disposed in the handle **70**, and the first shaft portion and the second shaft portion, which engage with the first hole portion and the second hole portion, may be disposed in the conveyance cover **50**.

In addition, although the description has been made for the image forming apparatus **1** in the present embodiment, the present invention is not limited to this. For example, the present invention may be applied to a large-capacity feeding deck that can be connected to the image forming apparatus **1**, or to another sheet conveyance apparatus, such as a post-process apparatus, that does not include the image forming portion **20**.

In addition, although the locking member **53** is disposed, in the present embodiment, so as to be able to slide in the vertical direction VD, the present disclosure is not limited to this. For example, the locking member **53** may be pivotally supported by the conveyance guide **52** or another member.

In addition, although the brake mechanism **106** is disposed in the side regulation plate **104** in the present embodiment, the present disclosure is not limited to this. For example, the brake mechanism **106** may be disposed in the side regulation plate **103** or a trailing-edge regulation plate that regulates the position of the trailing edge of the sheet.

In addition, although the brake mechanism **106** produces a predetermined frictional force, in the present embodiment, when the brake mechanism **106** moves toward the second direction WD2 in a state where the brake release lever **105** is not operated, the present disclosure is not limited to this. That is, the brake mechanism **106** may produce no frictional force when the brake mechanism **106** moves toward the second direction WD2, even in a state where the brake release lever **105** is not operated.

In addition, although the brake mechanism **106** is disposed, in the present embodiment, in the side regulation plate **104** of the manual-feed conveyance portion **2**, the present disclosure is not limited to this. For example, the brake mechanism **106** may be disposed in a side regulation plate disposed in the cassette **11**, or may be disposed in a side regulation plate disposed in an auto document feeder (ADF).

In addition, although the description has been made for the electrophotographic image forming apparatus **1** in the present embodiment, the present invention is not limited to this. For example, the present invention may also be applied to an ink-jet image forming apparatus that forms images on sheets by injecting ink from a nozzle.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary

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embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application No. 2022-091005, filed Jun. 3, 2022, which is hereby incorporated by reference herein in its entirety.

What is claimed is:

1. A sheet conveyance apparatus comprising:

an apparatus body including a conveyance assembly configured to convey a sheet; and
a door pivotally supported around a pivot axis disposed in a lower portion of the apparatus body such that the door is opened and closed with respect to the apparatus body, wherein the apparatus body includes a handle assembly including a first holder used to lift the apparatus body and is disposed inside a width of the door in an axis direction of the pivot axis,

wherein the handle assembly includes a first engaging section and a second engaging section disposed on the pivot axis,

wherein the door includes:

a first engaged section configured to engage with the first engaging section; and

a second engaged section configured to engage with the second engaging section, and

wherein the door is pivotally supported around the first engaging section and the second engaging section with respect to the handle assembly.

2. The sheet conveyance apparatus according to claim 1, wherein the first holder is disposed between the first engaging section and the second engaging section in the axis direction.

3. The sheet conveyance apparatus according to claim 1, wherein the first engaging section is disposed on a first end surface of the handle assembly in the axis direction, and

wherein the second engaging section is disposed on a second end surface of the handle assembly in the axis direction.

4. The sheet conveyance apparatus according to claim 3, wherein the first engaging section is disposed on a lower edge portion of the first end surface, and

wherein the second engaging section is disposed on a lower edge portion of the second end surface.

5. The sheet conveyance apparatus according to claim 1, wherein the first engaging section and the second engaging section are shaft portions extending in directions that extend away from each other in the axis direction, and

wherein the first engaged section and the second engaged section are hole portions configured to respectively engage with the first engaging section and the second engaging section.

6. The sheet conveyance apparatus according to claim 1, wherein a distance between the first engaging section and the second engaging section in the axis direction is equal to or larger than two thirds of a width of the door in the axis direction.

7. The sheet conveyance apparatus according to claim 1, wherein the door includes a conveyance guide configured to constitute a part of a conveyance path through which a sheet conveyed by the conveyance assembly passes, and

wherein a distance between the first engaging section and the second engaging section in the axis direction is larger than a length of the conveyance assembly in the axis direction.

8. The sheet conveyance apparatus according to claim 1, wherein the first holder is recessed with respect to an

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exterior surface of the door in a state where the door is closed with respect to the apparatus body.

9. The sheet conveyance apparatus according to claim 1, wherein the apparatus body includes a lockable assembly, wherein the door includes:

a second holder configured to be held by a user;
a conveyance assembly configured to constitute a part of a conveyance path through which a sheet conveyed by the conveyance assembly passes;

a cover movably supported by the door between a first position and a second position, the first position being a position at which the cover covers the second holder and is flush with an exterior surface of the door, the second position being a position that allows the user to access the second holder;

a lock configured to move with respect to the conveyance assembly between a locking position and an unlocking position, the locking position being a position at which the lock engages with the lockable assembly so as to lock the door on the apparatus body, the unlocking position being a position at which the lock is separated from the lockable assembly so as to allow the door to be opened and closed with respect to the apparatus body; and

a biasing element configured to urge the door toward a direction in which the door is opened with respect to the apparatus body,

wherein the lock moves from the locking position to the unlocking position in accordance with a movement of the cover from the first position toward the second position, and

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wherein the door is opened with respect to the apparatus body by an urging force of the biasing element in a case where the lock moves from the locking position to the unlocking position.

10. The sheet conveyance apparatus according to claim 9, wherein the lock moves from the locking position to the unlocking position by being pressed by the cover in a case where the cover moves from the first position toward the second position.

11. The sheet conveyance apparatus according to claim 9, wherein the lock is supported such that the lock is configured to slide with respect to the conveyance assembly between the locking position and the unlocking position.

12. The sheet conveyance apparatus according to claim 9, wherein the apparatus body includes a first roller, wherein the sheet conveyance apparatus further includes a second roller that is configured to be rotatably supported by the door, and that is in contact with the first roller in a state where the door is closed with respect to the apparatus body, and

wherein the biasing element is configured to urge the second roller toward the first roller in a state where the door is closed with respect to the apparatus body.

13. An image forming apparatus comprising:
the sheet conveyance apparatus according to claim 1; and
an image forming portion disposed in the apparatus body and configured to form an image on a sheet conveyed by the sheet conveyance apparatus.

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